

JOINT DISTRIBUTION IN RESIDUE CLASSES OF FAMILIES OF MULTIPLICATIVE FUNCTIONS I

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ABSTRACT. We study the joint distribution of families of multiplicative functions in residue classes, allowing the moduli to vary within a wide range and assuming some natural control on the average behavior of the functions at (some fixed power of) the primes. As applications, we obtain essentially best possible analogues of the Siegel–Walfisz theorem for families of multiplicative functions that can be controlled by polynomials at some powers of primes. (This class includes several interesting arithmetic functions such as Euler’s totient $\varphi(n)$, the sum-of-divisors $\sigma(n)$, and more generally, the sum-of-divisor-powers $\sigma_r(n) := \sum_{d|n} d^r$, and so on.) Our results (optimally) extend works of Narkiewicz, Rayner, Śliwa, Dobrowolski, Fomenko and others. One of the main ideas behind our arguments is the detection of a certain “mixing” (or “quantitative ergodicity”) phenomenon via methods from sieve theory and the anatomy of integers. Additionally, we combine ideas and machinery from classical analytic number theory, character sums, linear algebra over rings, as well as arithmetic and algebraic geometry.

1. INTRODUCTION

We say that a function $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ is **equidistributed mod $q \in \mathbb{Z}^+$** if $\#\{n \leq x : f(n) \equiv a \pmod{q}\} \sim x/q$ as $x \rightarrow \infty$. This notion has been studied for *additive* functions: See results of Delange [5, 6], which have been partially extended in [25, 26, 1, 31]. However, for *multiplicative* functions, this is not the correct notion to work with: For instance, using classical results (such as those implicit in [15]), it can be shown that Euler’s totient function $\varphi(n)$ is almost always divisible by any fixed integer q , and hence is **not** equidistributed modulo any fixed $q > 1$. Motivated by this, Narkiewicz defines a function $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ to be **weakly equidistributed (WUD) mod q** if there are infinitely many n for which $(f(n), q) = 1$, and if

$$\#\{n \leq x : f(n) \equiv a \pmod{q}\} \sim \frac{1}{\varphi(q)} \#\{n \leq x : (f(n), q) = 1\} \quad \text{as } x \rightarrow \infty,$$

for every coprime residue $a \pmod{q}$.¹ Hence, our sample space is $\{n : (f(n), q) = 1\}$, and every coprime residue mod q gets its fair share of the sample space. For example, if f is WUD mod 6, then the residues 1 and 5 mod 6 each asymptotically receive 50% of the sample space $\{n : (f(n), 6) = 1\}$. Likewise, given a family of functions $f_1, \dots, f_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$, we say that $(f_1, \dots, f_K)$ are **WUD mod q** if there are infinitely many n for which $(f_1(n) \dots f_K(n), q) = 1$, and if for all coprime residues $a_1, \dots, a_K \pmod{q}$, we have

$$(1.1) \quad \#\{n \leq x : (\forall i) f_i(n) \equiv a_i \pmod{q}\} \sim \frac{1}{\varphi(q)^K} \#\{n \leq x : (f_1(n) \dots f_K(n), q) = 1\} \quad \text{as } x \rightarrow \infty.$$

The notion of weak equidistribution has been studied by several authors. Narkiewicz gave a criterion [18, Theorem 1] deciding weak equidistribution of any “polynomially-defined” multiplicative function; here, $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ is **polynomially-defined** if there exist $V \in \mathbb{Z}^+$ and polynomials $\{F_v\}_{v=1}^V \subset \mathbb{Z}[T]$ satisfying $f(p^v) = F_v(p)$ for all primes p and all $v \in [V]$. (Hence, F_v controls the behavior of f at the v -th powers of primes.) As a consequence, he showed that $\varphi(n)$ is WUD mod q iff $(q, 6) = 1$. (Related are works of Dence–Pomerance [7] and Banks–Shparlinski [2, Theorem 3.1].) Śliwa [33] used Narkiewicz’s criterion to show that the sum-of-divisors $\sigma(n) = \sum_{d|n} d$ is WUD mod q iff $6 \nmid q$. This was extended to the functions

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¹Here, (a, b) denotes the gcd of a and b , unless stated otherwise.

$\sigma_r(n) = \sum_{d|n} d^r$ (which also appear as Fourier coefficients of Eisenstein series) by Narkiewicz, Rayner, Śliwa, Dobrowolski and Fomenko [33, 10, 23, 21, 22, 27, 28]. Finally, the most general results were also obtained by Narkiewicz in [20, 19], where he gave a complete characterization deciding when an arbitrary *family* of polynomially-defined multiplicative functions is weakly equidistributed to a fixed modulus.

In all these works, q is assumed to be *fixed*. It is then natural to ask what happens when q is allowed to *grow* with x , in analogy with the Siegel–Walfisz theorem from prime number theory. Analogues of Siegel–Walfisz are always sought both due to their independent interest and applications to other problems, and such analogues have been studied in other contexts, such as for smooth numbers and mean values of multiplicative functions. In our context, the explicit question is: Can we find analogues of the Siegel–Walfisz theorem for the joint distribution of families of multiplicative functions? To formalize this, given $f_1, \dots, f_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ and a set $\mathcal{Q} \subset \mathbb{Z}^+$, we say that $(f_1, \dots, f_K)$ are WUD (uniformly) mod $q \in \mathcal{Q}$, if:

- For every $q \in \mathcal{Q}$, we have $(\prod_{i=1}^K f_i(n), q) = 1$ for infinitely many n , and
- (1.1) holds as $x \rightarrow \infty$, uniformly in $q \in \mathcal{Q}$ and in coprime residues $a_1, \dots, a_K \bmod q$.

The only results in this direction seem to appear in [16, 24, 26], which make partial progress towards extending *highly special cases* of Narkiewicz’s criteria [18, 20] to varying q . In [16], Lebowitz-Lockard, Pollack and the author show that $\varphi(n)$ is WUD uniformly mod *prime* $q \in [5, (\log x)^{K_0}]$ for any fixed K_0 . In [24], Pollack and the author consider multiplicative $f_1, \dots, f_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ for which there exist nonconstant $F_1, \dots, F_K \in \mathbb{Z}[T]$ with the product $F_1 \dots F_K$ separable², such that $f_i(p) = F_i(p)$ for all i and all primes p . We show that such $(f_1, \dots, f_K)$ are WUD uniformly mod $q \leq (\log x)^{1/K-\epsilon}$ that are “almost prime” in a certain sense. Finally, [26] partially improves this only for $K = 1$, optimizing the range of q and replacing “almost prime” by the condition that q is supported on sufficiently large primes.

The above restrictions are unnecessary in the fixed- q results, but they cannot be overcome with the methods available in [16, 24, 26]. For instance, most arguments in [26] do not generalize to families (i.e. to $K > 1$), and the separability and the (roughness) condition on q are both strictly necessary for the arguments. Moreover, Narkiewicz’s criteria [18, 20] allow a certain flexibility with prime powers and the possibility of sparse input sets (see § 7.2), both of which are crucial in most of the other fixed- q results cited above; however [16, 24, 26] cannot allow any such flexibility. Hence, these works cannot give satisfactory uniform extensions of most of the other works of Narkiewicz, Rayner, Śliwa and others.

In this manuscript, we remove *all* these limitations and give essentially optimal uniform extensions of Narkiewicz’s most general criteria [20, 19] (for a single growing modulus), hence also completely extending all the cited works. Our q will vary either within optimal ranges *or* up to any fixed power of $\log x$, hence we also obtain essentially best possible qualitative analogues of the *Siegel–Walfisz theorem* for a general family of polynomially-defined multiplicative functions. To do this, we not only need to refine the methods in the previous works (which were restricted to the anatomy of integers), but also use several ideas not considered before, such as from linear algebra over rings, and from algebraic and arithmetic geometry. As mentioned in § 1.2, some of these ideas have potential applications to other related problems as well.

1.1. The Main Results. Since the general version of Narkiewicz’s criterion in [20] requires some technical set-up, we first give uniform analogues of a special case to illustrate the most important ideas. In section 7, we state our most general results and the additional ingredients they require.

Consider multiplicative functions $f_1, \dots, f_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ for which there exist polynomials $F_1, \dots, F_K \in \mathbb{Z}[T]$ satisfying $f_i(p) = F_i(p)$ for all primes p and all $i \in [K]$. For any $q \in \mathbb{Z}^+$, we define

$$\alpha(q) := \varphi(q)^{-1} \#\{u \in U_q : F_1(u) \dots F_K(u) \in U_q\}, \text{ where } U_q := (\mathbb{Z}/q\mathbb{Z})^\times.$$

Let $\mathcal{Q}(\widehat{f})$ be the set of all $q \in \mathbb{Z}^+$ with $\alpha(q) \neq 0$ that satisfy the following property:

²i.e. having no repeated roots in $\mathbb{C}$

For all Dirichlet characters $(\chi_1, \dots, \chi_K) \neq (\chi_0, \dots, \chi_0) \pmod q$ satisfying $\prod_{i=1}^K \chi_i(F_i(u)) = 1$ on its “unit support” $\{u \in U_q : \prod_{i=1}^K F_i(u) \in U_q\}$, we have $\prod_{i=1}^K \chi_i(f_i(2^j)) = -1$ for all $j \geq 1$.

Here χ_0 is the trivial character mod q . Narkiewicz’s criterion in this special setting is as follows.

Theorem 1.1. [20] *Assume $\alpha(q) \neq 0$. Then $(f_1, \dots, f_K)$ are WUD mod q if and only if $q \in \mathcal{Q}(\widehat{f})$.*

Now assume that $F_1, \dots, F_K$ are multiplicatively independent, i.e. $\prod_{i=1}^K F_i^{c_i}$ is nonconstant in $\mathbb{Q}(T)$ for all integers $(c_1, \dots, c_K) \neq (0, \dots, 0)$.³ Factor each F_i as $r_i \prod_{j=1}^M G_j^{\mu_{i,j}}$ with $r_i \in \mathbb{Z}$, $\mu_{i,j} \in \mathbb{Z}_{\geq 0}$, and with $\{G_j\}_{j=1}^M \subset \mathbb{Z}[T]$ being pairwise coprime primitive⁴ irreducibles, such that each G_j appears with a positive exponent $\mu_{i,j}$ in some F_i . Let E_0 be the $M \times K$ matrix $\left((\mu_{i,j})_{\substack{1 \leq i \leq K \\ 1 \leq j \leq M}} \right)^\top$, and $\beta \in \mathbb{Z}^+$ be the largest invariant factor of E_0 . Since $\{F_i\}_i$ are multiplicatively independent, the columns of E_0 are $\mathbb{Q}$ -linearly independent (i.e. $\text{rank}(E_0) = K \leq M$), so β is the last diagonal entry in the Smith Normal Form of E_0 .

Fix any $B_0 > 0$. We say that q satisfies Hypothesis $IFH(F_1, \dots, F_K; B_0)$ if all its prime divisors ℓ exceeding B_0 satisfy $\gcd(\ell - 1, \beta) = 1$. Note that if $\prod_{i=1}^K F_i$ is separable (as was assumed in previous works [24, 26]) then $\beta = 1$, so that any q satisfies $IFH(F_1, \dots, F_K; B_0)$ for any $B_0 > 0$.

Theorem 1.2. *Fix any $K_0, B_0 > 0$ and $\epsilon \in (0, 1)$, and assume that $F_1, \dots, F_K$ are multiplicatively independent. Then $(f_1, \dots, f_K)$ are WUD mod $q \leq (\log x)^{K_0}$ lying in $\mathcal{Q}(\widehat{f})$, such that q satisfies $IFH(F_1, \dots, F_K; B_0)$ along with at least one of the following two conditions:*

- (i) $q \leq (\log x)^{(1-\epsilon)\alpha(q)} (K-1/D_{\min})^{-1}$, where $D_{\min} := \min\{\deg F_i : 1 \leq i \leq K\}$, **or**
- (ii) q is squarefree and $q^{K-1} D_{\min}^{\omega(q)} \leq (\log x)^{(1-\epsilon)\alpha(q)}$.

For $K = D_{\min} = 1$ (single multiplicative function controlled by a linear polynomial at the primes), (i) and (ii) are tautological, so we have uniformity in $q \leq (\log x)^{K_0}$. Some applications of Theorem 1.2: $\varphi(n)$ is WUD mod $q \leq (\log x)^{K_0}$ coprime to 6 (extending Narkiewicz [18, Cor. 2]); $\sigma(n)$ is WUD mod odd $q \leq (\log x)^{K_0}$ (partially extending Śliwa [33]); (φ, σ) are WUD mod $q \leq (\log x)^{(1-\epsilon)\alpha_2(q)}$ coprime to 6, where $\alpha_2(q) = \prod_{\ell|q} (\ell - 3)/(\ell - 1)$ (extending Narkiewicz [19, Theorem 1] for a single modulus). A simple consequence of (ii) and (4.1) below is that if $F_1, \dots, F_K$ are multiplicatively independent, then $(f_1, \dots, f_K)$ are WUD mod prime $q \leq (\log x)^{(1/(K-1)-\epsilon)\alpha(q)}$ satisfying $q \in \mathcal{Q}(\widehat{f})$ and $\gcd(q - 1, \beta) = 1$.

In § 6, we construct $\{F_i\}_i$ to show that for any K and $D_{\min}$ (except $K = D_{\min} = 1$), both ranges in (i) and (ii) are essentially optimal, i.e. no $1 - \epsilon$ can be replaced by $1 + \epsilon$. We also show that weak equidistribution fails for $q > (\log x)^{(1+\epsilon)\alpha(q)/(K-1)}$ for any $F_i \in \mathbb{Z}[T]$. This also means that for $K > 1$, $D_{\min} = 1$, the ranges in (i) and (ii) are optimal even if we take *all* the F_i to be essentially arbitrary linear polynomials; in particular, the range $q \leq (\log x)^{(1-\epsilon)\alpha_2(q)}$ above is optimal even for the *particular example* of (φ, σ) . Moreover, the multiplicative independence hypothesis and hypothesis $IFH(F_1, \dots, F_K; B_0)$ are also both necessary in Theorem 1.2 and in Theorem 1.3 below: We shall establish this in a sequel paper [32].

As our constructions in section 6 show, obstructions to uniformity come from prime inputs. We can modify those constructions to produce more obstructions coming from inputs n that have *too few* large prime factors. Hence to restore uniformity in the full Siegel-Walfisz range, we need to restrict our n to those with *sufficiently many* large prime factors. To this end, let $\mathbf{P}_R(\mathbf{n})$ denote the R -th largest prime factor of n counted with multiplicity (with the convention $P_R(n) := 1$ if $\Omega(n) < R$).

³This hypothesis is easily satisfied in almost all applications: For example, it is satisfied if $\prod_{i=1}^K F_i$ is separable, or (more generally) if each F_i has an irreducible factor that is not present in the other F_j (for $j \neq i$).

⁴A polynomial in $\mathbb{Z}[T]$ is *primitive* when the greatest common divisor of its coefficients is 1.

Theorem 1.3. Fix $K_0, B_0 > 0$ and assume that $F_1, \dots, F_K$ are multiplicatively independent. Then

$$(1.2) \quad \sum_{\substack{n \leq x: P_R(n) > q \\ (\forall i) f_i(n) \equiv a_i \pmod{q}}} 1 \sim \frac{1}{\varphi(q)^K} \sum_{\substack{n \leq x: P_R(n) > q \\ (\prod_i f_i(n), q) = 1}} 1 \sim \frac{1}{\varphi(q)^K} \sum_{\substack{n \leq x \\ (\prod_i f_i(n), q) = 1}} 1 \quad \text{as } x \rightarrow \infty,$$

uniformly in $q \leq (\log x)^{K_0}$ with $q \in \mathcal{Q}(\hat{f})$ satisfying $IFH(F_1, \dots, F_K; B_0)$, and uniformly in $a_i \in U_q$; here

$$(1.3) \quad R = \begin{cases} KD + 1, & \text{in general, where } D := \sum_{i=1}^K \deg F_i, \\ 2K + 1, & \text{if } q \text{ is squarefree,} \\ 2, & \text{if } q \text{ squarefree, } K = 1, F_1 \text{ is not squarefull (i.e. } F_1 \text{ has a simple root in } \mathbb{C}). \end{cases}$$

For instance, (φ, σ) are WUD uniformly modulo $q \leq (\log x)^{K_0}$ coprime to 6 if we restrict to inputs n with $P_5(n) > q$. (Without this restriction, we only have the aforementioned optimal range $q \leq (\log x)^{(1-\epsilon)\alpha_2(q)}$.) Another application: By Narkiewicz [19, 20], $(\varphi, \sigma, \sigma_2)$ are WUD modulo any fixed q having $P^-(q) > 23$. By Theorem 1.2, this weak equidistribution holds uniformly modulo any such $q \leq (\log x)^{(1/2-\epsilon)c_q}$ where $c_q := \prod_{\substack{\ell|q \text{ prime} \\ \ell \equiv -1 \pmod{4}}} \frac{\ell-3}{\ell-1} \cdot \prod_{\substack{\ell|q \text{ prime} \\ \ell \equiv 1 \pmod{4}}} \frac{\ell-5}{\ell-1}$. Uniformity is restored modulo *all* $q \leq (\log x)^{K_0}$ with $P^-(q) > 23$, if we restrict to n with $P_{13}(n) > q$. This can be weakened to $P_7(n) > q$, if q is also assumed squarefree.

The smaller the value of R in (1.2), the larger our set wherein equidistribution occurs, and the better our result. In section 6, we show that the third value of R in (1.3) is optimal, while for *any* $K \in \mathbb{Z}^+$, the second value is at most “one step away” from being optimal (in that it cannot be lowered to $2K - 1$). Note that even the $K = 1$ special case of Theorem 1.3 improves over [26, Theorem 1.4]. It is not entirely clear whether the first value of R in (1.3) can be improved in the relatively special setting of Theorem 1.3 above; however, we remark that in the more general setting of Theorem 7.6 below, the corresponding value of R in (7.8) does turn to be optimal (as discussed after Theorem 7.6).

One important theme in our arguments is a “*mixing*” phenomenon originally observed in [26]: Given a nonconstant separable $G \in \mathbb{Z}[T]$ and q supported on large primes, if we choose uniformly at random, $u_1, u_2, \dots \in U_q$ such that each $G(u_j) \in U_q$, then in the sequence of partial products $\{G(u_1) \dots G(u_J)\}_{J=1}^\infty$, *every* unit mod q appears equally often. Here we detect and generalize this “mixing” via more careful arguments from sieve theory, the anatomy of integers and character sums (extensions of Weil bounds).

Moreover, we need several ideas and machinery not considered in previous works, such as from linear algebra over rings, and algebraic/arithmetical geometry (via regular sequences and counting rational points on varieties over finite fields). We also need a theorem of Halász (which requires us to carefully estimate certain “pretentious distances”⁵), and we need to *modify* the Landau–Selberg–Delange “LSD” method. (Known forms of LSD or other mean value results are *not* sufficient to get the desired uniformity in q .)

1.2. Beyond the polynomially-defined regime. The aforementioned “mixing”, extended suitably, can be used to study the joint weak equidistribution of multiplicative functions $h_1, \dots, h_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ assuming only a natural control on the *average* behavior of the h_i at the primes. Owing to this generality, we expect the following results to have applications beyond the “polynomially-defined” regime, such as for Ramanujan’s τ -function and Fourier coefficients of modular forms.

We assume that for some (large) *threshold parameter* y and for all Dirichlet characters $(\chi_1, \dots, \chi_K) \pmod{q}$, we have an estimate of the form $\sum_{y < p \leq Y} \prod_i \chi_i(h_i(p)) \approx \alpha_{\hat{\chi}} \sum_{y < p \leq Y} \prod_i \chi_0(h_i(p))$ for all $Y \geq y$, where $\alpha_{\hat{\chi}} \in \mathbb{C}$ depends only on $\hat{\chi} := (\chi_1, \dots, \chi_K)$. Assuming this with an essentially arbitrary error term, we bound the *extent* to which $(h_1, \dots, h_K)$ fails to be weakly equidistributed mod q . Our first result shows that for any $J \in \mathbb{Z}^+$ for which the J -th moment $\sum_{\chi_1, \dots, \chi_K} |\alpha_{(\chi_1, \dots, \chi_K)}|^J$ is small enough, $(h_1, \dots, h_K)$ are WUD mod q , upon restricting to inputs n with J many large prime factors (exceeding the *threshold* y).

⁵in the terminology of Granville and Soundararajan [11]

In what follows, we set $\widehat{\chi} := (\chi_1, \dots, \chi_K)$ and let $\mathfrak{f}(\widehat{\chi})$ denote the lcm of the conductors of χ_i . We write $\widehat{h} := (h_1, \dots, h_K)$, $h := h_1 \cdots h_K$, and $\widehat{\chi}(\widehat{h}(n)) := \chi_1(h_1(n)) \cdots \chi_K(h_K(n))$.

Theorem 1.4. *Let $h_1, \dots, h_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ be multiplicative functions. Consider any $q \in \mathbb{Z}^+$, $M \geq 1$, $y \geq q$, and a decreasing function $\mathcal{E} : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, such that for all $\widehat{\chi} \bmod q$ and all $Y \geq y$,*

$$(1.4) \quad \sum_{y < p \leq Y} \widehat{\chi}(\widehat{h}(p)) = \alpha_{\widehat{\chi}} \sum_{y < p \leq Y} \chi_0(h(p)) + O(MY\mathcal{E}(y)),$$

with $\alpha_{\widehat{\chi}} \in \mathbb{C}$ in the unit disk. Then for all d dividing q , all $a_i \in U_q$, all $x \geq y$ and all $J \in \mathbb{Z}^+$, we have

$$(1.5) \quad \sum_{\substack{n \leq x: P_J(n) > y \\ (\forall i) h_i(n) \equiv a_i \pmod{q}}} 1 - \left(\frac{\varphi(d)}{\varphi(q)} \right)^K \sum_{\substack{n \leq x: P_J(n) > y, (h(n), q) = 1 \\ (\forall i) h_i(n) \equiv a_i \pmod{d}}} 1 \\ \ll \frac{1}{\varphi(q)^K} \sum_{\substack{\widehat{\chi} \bmod q \\ \mathfrak{f}(\widehat{\chi}) \nmid d}} |\alpha_{\widehat{\chi}}|^J \sum_{\substack{n \leq x \\ (h(n), q) = 1}} 1 + \frac{Jx}{y} + Mx\mathcal{E}(y) \log x.$$

The implied constant here depends only on the implied constant in (1.4).

Note that (1.4) is in general weaker than a hypothesis like $\sum_{y < p \leq Y} \widehat{\chi}(\widehat{h}(p)) = \beta_{\widehat{\chi}} \sum_{y < p \leq Y} 1 + O(MY\mathcal{E}(y))$. Further, (1.5) just needs (1.4) for all $\widehat{\chi}$ with $\mathfrak{f}(\widehat{\chi}) \nmid d$. Taking $J = d = 1$ in (1.5) directly bounds the failure of weak equidistribution; however, to make the right side of (1.5) negligible in applications, we often need to select a suitable parameter $J \rightarrow \infty$. We thus see what happens for the n with $P_J(n) \leq y$.

Our estimates will be in terms of a parameter that controls the (suitably normalized) count of large prime solutions to the congruences $h_i(p) \equiv a_i \pmod{q}$. Given any $h_i : \mathbb{Z}^+ \rightarrow \mathbb{Z}$, any $q, r \in \mathbb{Z}^+$ and any $z > 0$, let $\xi_r(q; z) := \xi_r(q; z; \{h_i\}_{i=1}^K) > 0$ be *any* parameter satisfying the bound

$$(K_r) \quad \#\{P_1 \cdots P_r \leq x : P_i \text{ primes, } P_1 > z, q < P_r < \cdots < P_1, (\forall i) h_i(P_1) \cdots h_i(P_r) \equiv a_i \pmod{q}\} \\ \leq \xi_r(q; z) \cdot x(\log \log x)^r / \log(z/q)$$

for all $x \geq z$ and all $a_1, \dots, a_K \in U_q$. Thus, $\xi_r(q; z)$ controls the number of squarefree integers $A \leq x$ with exactly r prime factors (all suitably large), that satisfy $h_i(A) \equiv a_i \pmod{q}$ for all i . In particular, $\xi_1(q; z)$ controls the number of primes $P \in (z, x)$ satisfying $h_i(P) \equiv a_i \pmod{q}$ for all i . Note that this is genuinely a *definition*, not a *hypothesis*, for the choice $\xi_r(q; z) = 1$ always satisfies (K_r) by Chebyshev's estimates, for all possible $\{h_i\}_i, q, r$ and z . Let $\Psi(x, z)$ be the number of z -smooth numbers up to x .

Theorem 1.5. *Fix $K \in \mathbb{Z}^+$. Uniformly in **all** multiplicative functions $h_1, \dots, h_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$, all $q, J \in \mathbb{Z}^+$, all $a_i \in U_q$, and in all real numbers $x \geq z \geq y \geq q$, we have*

$$(1.6) \quad \sum_{\substack{n \leq x: P_J(n) \leq y \\ (\forall i) h_i(n) \equiv a_i \pmod{q}}} 1 \ll \xi_1(q; z) \frac{x(2 \log \log x)^{J+1}}{\log(z/q)} \exp \left(\sum_{p \leq y} \frac{\mathbb{1}_{(h(p), q) = 1}}{p} \right) + \Psi(x, z) + \frac{x}{y}.$$

$$(1.7) \quad \sum_{\substack{n \leq x: P_J(n) \leq y \\ (h(n), q) = 1}} 1 \ll \frac{x(2 \log \log x)^J}{\log z} \exp \left(\sum_{p \leq y} \frac{\mathbb{1}_{(h(p), q) = 1}}{p} \right) + \Psi(x, z) + \frac{x}{y}.$$

The implied constants depend only on K . For any $R \leq J$, if we also impose the condition $P_R(n) > q$ on the left of (1.6), then the same bound holds, only with $\xi_1(q; z)$ and $(2 \log \log x)^{J+1}$ replaced by

$$\xi_R(q; z) + R^K \sum_{\substack{1 \leq r < R \\ 1 \leq s < K}} \frac{\xi_r(q; z)}{q^{\max\{s, R-r-s\}}} \quad \text{and} \quad (2 \log \log x)^{J+R}, \quad \text{respectively.}$$

Combining Theorem 1.4 with Theorem 1.5, we get a bound the difference

$$\sum_{\substack{n \leq x: P_R(n) > q \\ (\forall i) h_i(n) \equiv a_i \pmod{q}}} 1 - \left(\frac{\varphi(d)}{\varphi(q)} \right)^K \sum_{\substack{n \leq x: (h(n), q) = 1 \\ (\forall i) h_i(n) \equiv a_i \pmod{d}}} 1$$

for all $d \mid q$, all $x, y, z > 0$, and all $J, R \in \mathbb{Z}^+$ satisfying $q \leq y \leq z \leq x$ and $J \geq R$. The flexibility in choosing d, y, z, J, R in this bound (especially the fact that z and J now appear only on the right of this bound) are very useful in applications: For instance, by taking $d = R = 1$ and a large J , we can make the right side negligible while including *all* inputs n on the left. The strength of this bound basically depends on the bounds available on $\sum_{f(\widehat{\chi}) \nmid d} |\alpha_{\widehat{\chi}}|^J$, and on $\xi_r(q; z)$ for $r \in [R]$. As discussed in Remark 2, all the “ $f(\widehat{\chi}) \nmid d$ ” conditions can be replaced by the more general “ $\widehat{\chi} \notin \mathcal{S}$ ”, with $\mathcal{S}$ an arbitrary subset of the set of K -tuples of characters mod q ; the corresponding variants then would be uniform in $\mathcal{S}$.

We also mention the following estimate, which forms a key input in the proofs of Theorems 1.2 and 1.3, and may also be of independent interest or have applications to other problems.

Proposition 1.6. *Under the hypotheses of Theorem 1.4, we have for any $\widehat{\chi} \pmod{q}$,*

$$\sum_{n \leq x} \widehat{\chi}(\widehat{h}(n)) \ll \frac{x}{\log x} \exp \left(\sum_{\substack{p \leq y \\ (h(p), q) = 1}} \frac{1}{p} + |\alpha_{\widehat{\chi}}| \sum_{\substack{y < p \leq x \\ (h(p), q) = 1}} \frac{1}{p} + O(M\mathcal{E}(y)(\log x)^3) \right).$$

The implied constant depends only on the implied constant in (1.4).

Finally, we remark that Narkiewicz’s weak equidistribution criteria [18, 20] are *much* more general than Theorem 1.1: They allow the possibility that the behavior of our f_i at the ν -th powers of primes is *most significant* (in a certain sense), for some fixed but arbitrary $\nu \in \mathbb{Z}^+$; this also allows the possibility of the input sets $\{n : \gcd(\prod_i f_i(n), q) = 1\}$ being sparse. However, all the results and methods from all previous work on growing moduli are unable to handle any $\nu > 1$. Adapting our methods for Theorems 1.2 and 1.3 with additional arguments, we obtain (in section 7) optimal uniform results in this greater generality, along with several new applications that could not be addressed with previous work on growing moduli. We also give extensions of the results in subsection § 1.2 that apply for certain sparse input sets.

In sections 2 to 5, we will first prove the results in subsection § 1.2, and then deduce those in § 1.1 with additional arguments. In a future paper, we shall study the joint weak equidistribution of a family of multiplicative functions $(f_1, \dots, f_K)$ to growing moduli $(q_1, \dots, q_K)$, defined appropriately.

Conventions. We do not consider the zero function as multiplicative (thus, $f(1) = 1$ for any multiplicative function f). By $P(n)$ and $P^-(n)$, we mean the largest and least prime factors of n respectively (with $P(1) := 1$, $P^-(1) := \infty$). By the z -smooth part (resp. z -rough part) of n , we mean the largest z -smooth (resp. z -rough) integer dividing n . When there is no danger of confusion, we write $(a_1, \dots, a_K)$ in place of $\gcd(a_1, \dots, a_K)$. Throughout, we use the letters p and ℓ to denote primes. Given a tuple of characters $\widehat{\psi} := (\psi_1, \dots, \psi_K) \pmod{q}$, we use $f(\psi_i)$ for the conductor of ψ_i , and we let $f(\widehat{\psi}) := \text{lcm}[f(\psi_1), \dots, f(\psi_K)]$.

Implied constants in $\ll$ and O -notation, and implicit constants in qualifiers like “sufficiently large”, may depend on any parameters declared as “fixed”. Other dependence will be noted explicitly (with, say, parentheses or subscripts). $\log_k$ denotes the k -th iterate of the natural logarithm.

2. THE GENERAL MIXING PHENOMENON: PROOF OF THEOREMS 1.4 AND 1.5

Proof of Theorem 1.4. We say that $n \leq x$ is “convenient” (or “conv”) if the J largest prime divisors of n exceed y and are distinct, i.e. if n can be uniquely written as $mP_J \cdots P_1$, where

$$(2.1) \quad L_m := \max\{y, P(m)\} < P_J < \cdots < P_1.$$

Now, the number of $n \leq x$ having a repeated prime factor exceeding y is at most $\sum_{p>y} \sum_{m \leq x/p^2} 1 \ll x/y$, which is absorbed in the right of (1.5). Hence, to complete the proof of Theorem 1.4, it suffices to show (1.5) with the conditions “ $P_J(n) > y$ ” replaced by the conditions that n is convenient.

Using the orthogonality of Dirichlet characters, we detect the congruences $h_i(n) \equiv a_i \pmod{q}$ to get

$$(2.2) \quad \sum_{\substack{n \leq x \text{ conv} \\ (\forall i) \ h_i(n) \equiv a_i \pmod{q}}} 1 = \frac{1}{\varphi(q)^K} \sum_{\hat{\chi} \bmod q} \bar{\chi}_1(a_1) \cdots \bar{\chi}_K(a_K) \sum_{n \leq x \text{ conv}} \hat{\chi}(\hat{h}(n)).$$

For any $\hat{\chi} \bmod q$ for which $\mathfrak{f}(\hat{\chi}) \mid d$, there exists a unique tuple of characters $\hat{\psi} := (\psi_1, \dots, \psi_K) \bmod d$ such that $\chi_i = \chi_0 \psi_i$ for all i . Thus the total of contribution of all such $\hat{\chi}$ to the right of (2.2) equals

$$(2.3) \quad \frac{1}{\varphi(q)^K} \sum_{\hat{\psi} \bmod d} \bar{\psi}_1(a_1) \cdots \bar{\psi}_K(a_K) \sum_{\substack{n \leq x \text{ conv} \\ (h(n), q) = 1}} \hat{\psi}(\hat{h}(n)) = \left(\frac{\varphi(d)}{\varphi(q)} \right)^K \sum_{\substack{n \leq x \text{ conv: } (h(n), q) = 1 \\ (\forall i) \ h_i(n) \equiv a_i \pmod{d}}} 1.$$

Now consider any $\hat{\chi} \bmod q$. Splitting any convenient n uniquely as $mP_J \cdots P_1$ above, we obtain

$$(2.4) \quad \sum_{n \leq x \text{ conv}} \hat{\chi}(\hat{h}(n)) = \sum_{m \leq x} \hat{\chi}(\hat{h}(m)) \cdot \frac{1}{J!} \sum_{\substack{P_1, \dots, P_J > L_m \text{ distinct} \\ P_1 \cdots P_J \leq x/m}} \hat{\chi}(\hat{h}(P_1)) \cdots \hat{\chi}(\hat{h}(P_J)),$$

where we have replaced the ordering on $P_1, \dots, P_J$ in (2.1) by the weaker condition “ $P_1, \dots, P_J$ distinct” at the cost of the $J!$ factor. For each $i \in [J]$, we see by (1.4) that

$$(2.5) \quad \sum_{\substack{P_i \leq x/mP_1 \cdots P_{i-1}P_{i+1} \cdots P_J \\ P_i > L_m, \ r \neq i \implies P_r \neq P_i}} \hat{\chi}(\hat{h}(P_i)) = \sum_{L_m < P_i \leq x/mP_1 \cdots P_{i-1}P_{i+1} \cdots P_J} \hat{\chi}(\hat{h}(P_i)) + O(J) \\ = \alpha_{\hat{\chi}} \sum_{\substack{P_i \leq x/mP_1 \cdots P_{i-1}P_{i+1} \cdots P_J \\ P_i > L_m, \ r \neq i \implies P_r \neq P_i}} \chi_0(h(P_i)) + O\left(J + \frac{Mx\mathcal{E}(y)}{mP_1 \cdots P_{i-1}P_{i+1} \cdots P_J}\right),$$

The sum of the O -term over all m and all the other P_r in (2.4) is $\ll x/y + J^{-1}Mx\mathcal{E}(y) \log x$. (To see this, we use the $(J-1)!$ from (2.4) to restore the ordering on $P_1, \dots, P_{i-1}, P_{i+1}, \dots, P_J$, and then note that $mP_1 \cdots P_{i-1}P_{i+1} \cdots P_J \leq x/y$.) Hence using (2.5) for $i = 1, \dots, J$ successively, we get from (2.4),

$$(2.6) \quad \sum_{n \leq x \text{ conv}} \hat{\chi}(\hat{h}(n)) = (\alpha_{\hat{\chi}})^J \sum_{m \leq x} \hat{\chi}(\hat{h}(m)) \sum_{\substack{P_1, \dots, P_J \\ P_1 \cdots P_J \leq x/m \\ L_m < P_J < \cdots < P_1}} \chi_0(h(P_1 \cdots P_J)) + O\left(\frac{Jx}{y} + Mx\mathcal{E}(y) \log x\right).$$

The first sum on the right side is $\ll |\alpha_{\hat{\chi}}|^J \#\{n \leq x \text{ conv} : (h(n), q) = 1\}$. Using this bound for all character tuples $\hat{\chi} \bmod q$ for which $\mathfrak{f}(\hat{\chi}) \nmid d$, and combining this with (2.3), we obtain Theorem 1.4. $\square$

Proof of Theorem 1.5. We first show the analogue of (1.6) for a general R . By the argument in the sentence after (2.1), we see that the n that are either z -smooth or have a repeated prime factor exceeding y contribute $O(\Psi(x, z) + x/y)$. Let $\sum_{n \leq x}^*$ denote a sum over $n \leq x$ with a squarefree y -rough part, that

satisfy $P_R(n) > q$, $P_J(n) \leq y$, $P(n) > z$, and $h_i(n) \equiv a_i \pmod{q}$ for all i . Let $E_{r,s}$ be the number of n counted in $\sum_{n \leq x}^* 1$ with $\#\{p > q : p \parallel n\} = r$ and $\#\{p > q : p^2 \mid n\} = s$. Then

$$(2.7) \quad \sum_{n \leq x}^* 1 \leq \sum_{\substack{n \leq x \\ \#\{p > q : p \parallel n\} \geq R}}^* 1 + \sum_{\substack{n \leq x \\ \#\{p > q : p^2 \mid n\} \geq K}}^* 1 + \sum_{\substack{1 \leq r < R \\ 1 \leq s < K}} E_{r,s}.$$

Any n counted in $E_{r,s}$ is of the form $mp_1^{c_1} \cdots p_s^{c_s} A$, where $P(m) \leq q$, all $p_j > q$ are primes, all $c_j \geq 2$, where A is squarefree with $\Omega(A) = r$, $P^-(A) > q$, $P(A) > z$, and where all the factors are pairwise coprime. Thus $(h(m), q) = 1$, $h_i(A) \equiv a_i h_i(mp_1^{c_1} \cdots p_s^{c_s})^{-1} \pmod{q}$, and $\sum_{j=1}^s c_j \geq R - r$ (as $P_R(n) > q$). By (K_r) , the number of possible A is $\ll \xi_r(q; z) x (\log_2 x)^r / (mp_1^{c_1} \cdots p_s^{c_s} \log(z/q))$. Next,

$$(2.8) \quad \sum_{\substack{m: P(m) \leq q \\ (h(m), q) = 1}} \frac{1}{m} \leq \prod_{p \leq q} \left(1 + \sum_{r \geq 1} \frac{\mathbb{1}_{(h(p^r), q) = 1}}{p^r} \right) \ll \exp \left(\sum_{p \leq q} \frac{\mathbb{1}_{(h(p), q) = 1}}{p} \right),$$

and $\sum_{p_1, \dots, p_s > q} p_1^{-c_1} \cdots p_s^{-c_s} \ll q^{-(c_1 + \cdots + c_s - s)}$. The sum of $q^{-(c_1 + \cdots + c_s - s)}$ over all possible c_i is both $\ll q^s \prod_{j=1}^s (\sum_{c_j \geq 2} q^{-c_j}) \ll q^{-s}$, and $\ll \sum_{t \geq R-r} t^s / q^{t-s} \ll R^K / q^{R-r-s}$ (taking $t = \sum_{j=1}^s c_j$). Hence

$$E_{r,s} \ll R^K \frac{\xi_r(q; z)}{q^{\max\{s, R-r-s\}}} \cdot \frac{x (\log_2 x)^R}{\log(z/q)} \exp \left(\sum_{p \leq q} \frac{\mathbb{1}_{(h(p), q) = 1}}{p} \right).$$

The other two sums on the right of (2.7) are bounded analogously: For example, to bound the second sum, we write $n = mp_1^{c_1} \cdots p_K^{c_K} P$ with $P_J(m) \leq y$, $P > z$, and all $p_j > q$, $c_j \geq 2$. By Chebyshev's estimates, the count of P is $\ll x / (mp_1^{c_1} \cdots p_K^{c_K} \log z)$. This time in place of (2.8), we use

$$(2.9) \quad \sum_{\substack{m \leq x: P_J(m) \leq y \\ (h(m), q) = 1}} \frac{1}{m} \leq \sum_{\substack{A \leq x \\ \Omega(A) \leq J}} \frac{1}{A} \sum_{\substack{B: P(B) \leq y \\ (h(B), q) = 1}} \frac{1}{B} \ll (2 \log_2 x)^J \exp \left(\sum_{p \leq y} \frac{\mathbb{1}_{(h(p), q) = 1}}{p} \right),$$

where we have written $m = AB$ with $P(B) \leq y < P^-(A)$ (so that $P_J(n) \leq y$ forces $\Omega(A) \leq J$), bounded $\sum 1/B$ analogously to (2.8), and noted that $\sum 1/A \leq (1 + \sum_{p \leq x} 1/p)^J$. This establishes (1.6).

To show (1.7), we again impose the conditions $P(n) > z$ and that n has a squarefree y -rough part. Write any such $n = mP$ with $\max\{z, P(m)\} < P \leq x/m$, so that $P_J(m) \leq y$ and $(h(m), q) = 1$. By Chebyshev's estimates, the number of P is $\ll x / (m \log z)$. Bounding $\sum 1/m$ by (2.9) yields (1.7). $\square$

Remark. Given any set $\mathcal{S}$ of K -tuples of characters mod q , applying (2.6) for $\hat{\chi} \notin \mathcal{S}$, we obtain a variant of (1.5) with the condition $\mathfrak{f}(\hat{\chi}) \nmid d$ replaced by $\hat{\chi} \notin \mathcal{S}$, and with the second term on the left replaced by $\varphi(q)^{-K} \sum_{\hat{\chi} \in \mathcal{S}} \bar{\chi}_1(a_1) \cdots \bar{\chi}_K(a_K) \sum_{n \leq x: P_J(n) > y} \hat{\chi}(\hat{h}(n))$. This estimate is uniform in all $\mathcal{S}$.

3. A "HALASZ-TYPE" CHARACTER SUM BOUND: PROOF OF PROPOSITION 1.6

Consider any $\hat{\chi} \pmod{q}$. By Halász's Theorem (as stated in [34, Corollary III.4.12]), we have

$$(3.1) \quad \sum_{n \leq x} \hat{\chi}(\hat{h}(n)) \ll \frac{x}{L} + x \exp \left(- \min_{-L \leq t \leq L} \mathcal{D}(x, t) \right),$$

uniformly in $x, L \geq 2$, where by Mertens' Theorem,

$$(3.2) \quad \mathcal{D}(x, t) := \sum_{p \leq x} \frac{1 - \operatorname{Re}(\hat{\chi}(\hat{h}(p)) p^{-it})}{p} \geq \log_2 x - \sum_{\substack{p \leq y \\ (h(p), q) = 1}} \frac{1}{p} - \left| \sum_{y < p \leq x} \frac{\hat{\chi}(\hat{h}(p))}{p^{1+it}} \right| + O(1).$$

Set $L := \log x$. Cover $(y, x]$ with disjoint subintervals $I_\eta := (\eta, \eta(1 + 1/\log^2 x)]$, such that the rightmost of these may jut out past x but not past $x(1 + 1/\log^2 x)$. Uniformly in $p \in I_\eta$ and $t \in [-L, L]$, we have $p = \eta(1 + O(1/\log^2 x))$, so that $p^{-(1+it)} = \eta^{-(1+it)}(1 + O(1/\log x))$. Combining these observations with the estimates $\sum_\eta \sum_{p \in I_\eta} 1/\eta \ll \sum_{p \leq 2x} 1/p \ll \log_2 x$ and $\sum_{x < p \leq x(1+1/\log^2 x)} 1/p \ll 1/\log^3 x$, we get

$$(3.3) \quad \sum_{y < p \leq x} \frac{\widehat{\chi}(\widehat{h}(p))}{p^{1+it}} = \sum_\eta \frac{1}{\eta^{1+it}} \sum_{p \in I_\eta} \widehat{\chi}(\widehat{h}(p)) + O\left(\frac{\log_2 x}{\log x}\right).$$

Using (1.4) to estimate $\sum_{p \in I_\eta} \widehat{\chi}(\widehat{h}(p))$ and noting that there are $O(\log^3 x)$ many I_η 's, we obtain

$$\sum_{y < p \leq x} \frac{\widehat{\chi}(\widehat{h}(p))}{p^{1+it}} = \alpha_{\widehat{\chi}} \sum_\eta \frac{1}{\eta^{1+it}} \sum_{p \in I_\eta} \chi_0(h(p)) + O\left(\frac{\log_2 x}{\log x} + M\mathcal{E}(y)(\log x)^3\right).$$

Using the observations before (3.3), we can replace the double sum above by $\alpha_{\widehat{\chi}} \sum_{y < p \leq x} \chi_0(h(p))/p^{1+it}$ upto an error $\ll \log_2 x / \log x$. Inserting the resulting estimate into (3.2), we obtain

$$(3.4) \quad \mathcal{D}(x, t) \geq \log_2 x - \sum_{\substack{p \leq y \\ (h(p), q)=1}} \frac{1}{p} - |\alpha_{\widehat{\chi}}| \sum_{\substack{y < p \leq x \\ (h(p), q)=1}} \frac{1}{p} + O(1 + M\mathcal{E}(y)(\log x)^3).$$

Finally, inserting this bound into (3.1) completes the proof of Proposition 1.6. $\square$

4. SETTING THE STAGE: CHARACTER SUMS, MODULE THEORY, ARITHMETIC GEOMETRY

Until section 6, we assume the set up of Theorems 1.2 and 1.3, and let $\mathbf{f} = \prod_{i=1}^K \mathbf{f}_i$, $\mathbf{F} = \prod_{i=1}^K \mathbf{F}_i$. Since $\alpha(q) > 0$, and $\alpha(q) = \prod_{\ell|q} \alpha(\ell)$ by the Chinese Remainder Theorem, we have $\alpha(\ell) > 0$ for each $\ell \mid q$. Hence $\alpha(\ell) = 1 - \frac{1}{\ell-1} \#\{u \in U_\ell : F(u) \equiv 0 \pmod{\ell}\} \geq 1 - \frac{\min\{\ell-2, D\}}{\ell-1}$. By Mertens' Theorem,

$$(4.1) \quad \alpha(q) \gg (\log_2(3q))^{-D} \text{ uniformly in } q \geq 1 \text{ s.t. } \alpha(q) \neq 0, \text{ where } D = \deg F = \sum_{i=1}^K \deg F_i.$$

For an odd prime power ℓ^e , let ψ_{ℓ^e} denote a character mod ℓ^e which generates the character group of U_{ℓ^e} . We will need the following version of the Weil bounds, as in [35, Corollary 2.3].

Proposition 4.1. *Let ℓ be a prime. Consider any $H \in \mathbb{Z}[T]$ which is not of the form $c \cdot G^{\ell-1}$ in $\mathbb{F}_\ell[T]$. Then $|\sum_{u \bmod \ell} \psi_\ell(H(u))| \leq (r-1)\sqrt{\ell}$ where r is the number of distinct roots of H (in $\mathbb{C}$).*

We will also need Cochrane's [4, Theorems 1.2, 7.1; eq. (1.15)] extension of this to higher prime powers.

Proposition 4.2. *Let ℓ be a prime, $H \in \mathbb{Z}[T]$ be nonconstant with $\text{ord}_\ell(H) = 0$. Define $t = \text{ord}_\ell(H')$ and $M = \max\{\text{mult}_\theta(\ell^{-t}H') : \theta \in \mathbb{F}_\ell, H(\theta) \neq 0\}$. Let $e \geq t+2$ be an integer.*

- (i) *If $\ell > 2$, then $\left| \sum_{u \bmod \ell^e} \psi_{\ell^e}(H(u)) \right| \leq \left(\sum_{\substack{\theta \in \mathbb{F}_\ell \\ H(\theta) \neq 0}} \text{mult}_\theta(\ell^{-t}H') \right) \ell^{t/(M+1)} \ell^{e(1-1/(M+1))}$.*
- (ii) *Assume that $e \geq t+3$ and let ψ be the character mod 2^e defined by $\psi(5) = \exp(2\pi i/2^{e-2})$ and $\psi(-1) = 1$. Then $\left| \sum_{u \bmod 2^e} \psi(H(u)) \right| \leq (12.5) 2^{t/(M+1)} 2^{e(1-1/(M+1))}$.*

The non-constantness of $H \in \mathbb{Z}[T]$ guarantees that $t = \text{ord}_\ell(H')$ is finite, so that $\ell^{-t}H'$ can be viewed as a nonzero element of $\mathbb{F}_\ell[T]$. Hence, $\text{mult}_\theta(\ell^{-t}H') \in \mathbb{Z}_{\geq 0}$ is the multiplicity of $\theta \in \mathbb{F}_\ell$ as a zero of $\ell^{-t}H'$.

The following result controls the quantities appearing in the above bounds.

Lemma 4.3. *There exists a constant $C := C(F_1, \dots, F_K) > 0$ satisfying all the following:*

- (1) *For any prime $\ell > C$ satisfying $\gcd(\ell - 1, \beta) = 1$, the only tuple $(A_1, \dots, A_K) \in [\ell - 1]^K$ for which $\prod_{i=1}^K F_i^{A_i}$ is of the form $c \cdot G^{\ell-1}$ in $\mathbb{F}_\ell[T]$ is $(A_1, \dots, A_K) = (\ell - 1, \dots, \ell - 1)$.*
- (2) *$\text{ord}_\ell(\sum_{i=1}^K A_i F'_i \prod_{j \neq i} F_j) \leq \mathbb{1}_{\ell \leq C} C$ for any prime ℓ and $A_1, \dots, A_K \in \mathbb{Z}^+$ with $\ell \nmid \gcd(A_1, \dots, A_K)$.*
- (3) *Consider any integer $r \geq 2$, any prime ℓ , and any $A_1, \dots, A_K \in \mathbb{Z}^+$ with $\ell \nmid \gcd(A_1, \dots, A_K)$. Let $H(T) := T^{\varphi(\ell^r)} \prod_{i=1}^K F_i(T)^{A_i}$, $\tau_\ell := \text{ord}_\ell(H')$, and $\tilde{F}(T) := \sum_{i=1}^K A_i F'_i(T) \prod_{j \neq i} F_j(T)$. Then*

$$(4.2) \quad \tau_\ell = \begin{cases} \text{ord}_\ell(\tilde{F}) = 0, & \text{if } \ell > C, \text{ and } r \geq 2 \\ \text{ord}_\ell(\tilde{F}) \leq C, & \text{if } \ell \leq C, \text{ ord}_\ell(F) = 0, \text{ and } r \geq C + 2. \end{cases}$$

In both cases above, we have $\text{mult}_\theta(\ell^{-\tau_\ell} H') = \text{mult}_\theta(\ell^{-\tau_\ell} \tilde{F})$ for any $\theta \in \mathbb{F}_\ell$ for which $\theta F(\theta) \neq 0$, so that $\sum_{\theta \in \mathbb{F}_\ell: H(\theta) \neq 0} \text{mult}_\theta(\ell^{-\tau_\ell} H') = \sum_{\theta \in \mathbb{F}_\ell: \theta F(\theta) \neq 0} \text{mult}_\theta(\ell^{-\tau_\ell} \tilde{F}) \leq \deg(\tilde{F}) \leq D - 1$.

Proof. Write each $F'_i \prod_{j \neq i} F_j =: \sum_{r=0}^{D-1} c_{i,r} T^r$ in $\mathbb{Z}[T]$. Let $\tilde{\beta} \in \mathbb{Z}^+$ be the largest invariant factor of the $D \times K$ matrix $M_0 := \left((c_{i,r})_{\substack{1 \leq i \leq K \\ 0 \leq r < D}} \right)^\top$. Fix $C > \max \left\{ 2, \tilde{\beta}, \prod_{j=1}^M |\text{disc}(G_j)| \cdot \prod_{1 \leq j \neq j' \leq M} |\text{res}(G_j, G_{j'})| \right\}$, such that C exceeds the sizes of the leading coefficients of all F_i and G_j . (Here $\{G_j\}_{j=1}^M \subset \mathbb{Z}[T]$ are as before Theorem 1.2; disc and res are the discriminant and resultant.) We show that any such C satisfies all the assertions. Note that $\prod_{j=1}^M G_j$ is separable over $\mathbb{F}_\ell$ (i.e. squarefree in $\overline{\mathbb{F}_\ell}[T]$) for all $\ell > C$.

Proof of (1). Consider any prime $\ell > C$ and any $(A_1, \dots, A_K) \in [\ell - 1]^K$ satisfying $\gcd(\ell - 1, \beta) = 1$ and $\prod_{i=1}^K F_i^{A_i} = c \cdot G^{\ell-1}$ for some $c \in U_\ell$ and $G \in \mathbb{F}_\ell[T]$. Since each $F_i = r_i \prod_{j=1}^M G_j^{\mu_{i,j}}$, we have $c \cdot G^{\ell-1} = \rho \prod_{j=1}^M G_j^{\sum_{i=1}^K \mu_{i,j} A_i}$ for some $\rho \in U_\ell$. Factoring out all leading coefficients and comparing exponents, we get $\sum_{i=1}^K \mu_{i,j} A_i \equiv 0 \pmod{\ell - 1}$ for all $j \in [M]$ (as $\prod_{j=1}^M G_j$ is squarefree in $\overline{\mathbb{F}_\ell}[T]$).

This gives the matrix congruence $E_0(A_1 \dots A_K)^\top \equiv 0 \pmod{\ell - 1}$. Let $\text{diag}(\beta_1, \dots, \beta_K)$ be the Smith normal form of E_0 , so each $\beta_i \in \mathbb{Z}^+$, $\beta_i \mid \beta_{i+1}$, and $\beta = \beta_K$. Consider invertible integer matrices P_0, R_0 s.t. $P_0 E_0 R_0 = \text{diag}(\beta_1, \dots, \beta_K)$. The last matrix congruence yields $\text{diag}(\beta_1, \dots, \beta_K) R_0^{-1} (A_1 \dots A_K)^\top \equiv 0 \pmod{\ell - 1}$. Since $\gcd(\ell - 1, \beta_K) = 1$, we have $\gcd(\ell - 1, \beta_1 \dots \beta_K) = 1$, so that writing out the last congruence elementwise, we get $R_0^{-1} (A_1 \dots A_K)^\top \equiv 0 \pmod{\ell - 1}$. Hence $(A_1 \dots A_K)^\top \equiv 0 \pmod{\ell - 1}$.

Proof of (2, 3). Since $(\prod_{i=1}^K F_i^{b_i})' = \sum_{i=1}^K b_i F'_i \prod_{j \neq i} F_j$ for any $\{b_i\}_{i=1}^K \subset \mathbb{Q}$, the multiplicative independence of $\{F_i\}_{i=1}^K$ forces the $\mathbb{Q}$ -linear independence of $\left\{ F'_i \prod_{j \neq i} F_j \right\}_{i=1}^K$. Hence the matrix M_0 above has rank K ($\leq D$), its Smith Normal Form has K positive diagonal entries, and $\tilde{\beta}$ is the last diagonal entry.

Let $\tilde{F} := \sum_{i=1}^K A_i F'_i \prod_{j \neq i} F_j$ and $d := \text{ord}_\ell(\tilde{F})$. Since $\tilde{F}(T) = \sum_{r=0}^{D-1} (\sum_{i=1}^K c_{i,r} A_i) T^r$, we have the matrix congruence $M_0(A_1 \dots A_K)^\top \equiv 0 \pmod{\ell^d}$. An argument analogous to that given in subpart (1) shows that if $\ell^d \nmid \tilde{\beta}$, then $\ell \mid (A_1, \dots, A_K)$, a contradiction. Thus $\ell^d \mid \tilde{\beta}$, i.e. $d \leq v_\ell(\tilde{\beta}) \leq \mathbb{1}_{\ell \leq C} C$, proving (2).

Now (4.2) follows from the identity $H'(T) = \varphi(\ell^r) T^{\varphi(\ell^r)-1} \prod_{i=1}^K F_i(T)^{A_i} + T^{\varphi(\ell^r)} \tilde{F}(T) \prod_{i=1}^K F_i(T)^{A_i-1}$. This identity also shows that in both cases of (4.2), we have $\ell^{-\tau_\ell} H'(T) \equiv T^{\varphi(\ell^r)} \cdot \ell^{-\tau_\ell} \tilde{F}(T) \cdot \prod_{i=1}^K F_i(T)^{A_i-1}$ in the ring $\mathbb{F}_\ell[T]$, which establishes the remaining assertion of subpart (3) of the lemma. $\square$

For any tuple of characters $\hat{\chi} = (\chi_1, \dots, \chi_K) \pmod{q}$, we define

$$(4.3) \quad \alpha_{\hat{\chi}} = (\alpha(q) \varphi(q))^{-1} \sum_{u \in U_q} \hat{\chi}(\hat{F}(u)) = (\alpha(q) \varphi(q))^{-1} \sum_{u \pmod{q}} \chi_0(u) \prod_{i=1}^K \chi_i(F_i(u)).$$

With B_0 and D as in Theorems 1.2 and 1.3, and with C as in the proof of Lemma 4.3,

$$(4.4) \quad \text{Fix } C_0 > \max\{B_0, C, (32D)^{4D+2}\}, \text{ and then fix an integer } \kappa > 100D(DC_0^{2C_0})^{4C_0}.$$

Let $\mathbf{Q}_0$ be the largest C_0 -smooth $(\kappa + 1)$ -free divisor of q , i.e. $\mathbf{Q}_0 = \prod_{\ell \leq C_0} \ell^{\min\{v_\ell(q), \kappa\}} \ll 1$.

Proposition 4.4. (1) We have $\sum_{\substack{\widehat{\chi} \bmod q \\ \mathfrak{f}(\widehat{\chi}) \nmid \mathbf{Q}_0}} |\alpha_{\widehat{\chi}}|^N = o(1)$ as $N \rightarrow \infty$, uniformly in all q .

(2) Bound (K_r) holds with $h_i = f_i$, where uniformly in all $q \in \mathbb{Z}^+$ and $z \in (q, x]$, we have

$$(4.5) \quad \xi_r(q; z) \ll \begin{cases} \varphi(q)^{-K} \exp(O((\log q)^{1-1/D})), & \text{for each fixed } r \geq KD + 1. \\ q^{-r/D+\epsilon}, & \text{for each fixed } r \leq KD \text{ and any fixed } \epsilon > 0. \\ q^{-1/D_{\min}} \cdot \log_2(3q), & \text{if } r = 1, \text{ where } D_{\min} = \min_i \deg(F_i). \end{cases}$$

(3) Bound (K_r) holds with $h_i = f_i$, where uniformly in all **squarefree** q and $z \in (q, x]$, we have

$$(4.6) \quad \xi_r(q; z) \ll \begin{cases} \varphi(q)^{-K} \exp(O(\sqrt{\log q})), & \text{for each fixed } r \geq 2K + 1. \\ q^{-r/2+\epsilon}, & \text{for each fixed } r \leq 2K \text{ and any fixed } \epsilon > 0. \\ D_{\min}^{\omega(q)} \cdot \varphi(q)^{-1}, & \text{if } r = 1. \\ \varphi(q)^{-1} \exp(O(\sqrt{\log q})), & \text{if } r = 2, K = 1, \text{ and } F_1 \text{ is not squarefull.} \end{cases}$$

Proof. As seen before (4.1), $\alpha(\ell) \geq 1 - \frac{\min\{\ell-2, D\}}{\ell-1} > 0$ for each $\ell \mid q$, which gives both $\text{ord}_\ell(F) = 0$, and

$$(4.7) \quad \alpha(\ell) \geq \frac{1}{2} \text{ if } \ell > C_0; \quad \alpha(\ell) \geq \frac{1}{C_0} \text{ if } \ell \leq C_0; \quad \alpha(2) = 1 \text{ if } 2 \mid q.$$

We assume in the entire proof that $D > 1$; the case $D = 1$ (i.e. $K = 1$ and F_1 linear) is much simpler. We first claim that for any tuple $\widehat{\chi} = (\chi_1, \dots, \chi_K)$ of Dirichlet characters mod q , we have

$$(4.8) \quad |\alpha_{\widehat{\chi}}| \leq \prod_{\substack{\ell > C_0 \\ \ell \mid \mathfrak{f}(\widehat{\chi})}} \left(4D \cdot \ell^{-v_\ell(\mathfrak{f}(\widehat{\chi}))/D} \right) \prod_{\substack{\ell \leq C_0 \\ v_\ell(\mathfrak{f}(\widehat{\chi})) \geq \kappa+1}} \left(2DC_0^{C_0+1} \cdot \ell^{-v_\ell(\mathfrak{f}(\widehat{\chi}))/D} \right).$$

To show this, we start by writing each $\chi_i = \prod_{\ell^e \parallel q} \chi_{i,\ell}$ for some character $\chi_{i,\ell} \bmod \ell^e$. Then with $\alpha_{\widehat{\chi},\ell} := (\alpha(\ell)\varphi(\ell^e))^{-1} \sum_{u \bmod \ell^e} \chi_{0,\ell}(u) \prod_{i=1}^K \chi_{i,\ell}(F_i(u))$, we have $\alpha_{\widehat{\chi}} = \prod_{\ell^e \parallel q} \alpha_{\widehat{\chi},\ell}$ by the Chinese Remainder Theorem. Now let $r_\ell := v_\ell(\mathfrak{f}(\widehat{\chi}))$, so $\ell^{r_\ell} = \text{lcm}[\mathfrak{f}(\chi_{1,\ell}), \dots, \mathfrak{f}(\chi_{K,\ell})]$. Let $\psi_{i,\ell}$ denote the character mod ℓ^{r_ℓ} inducing $\chi_{i,\ell}$, so that **at least one of $(\psi_{1,\ell}, \dots, \psi_{K,\ell})$ is primitive mod ℓ^{r_ℓ}** . Note that

$$(4.9) \quad \alpha_{\widehat{\chi},\ell} = (\alpha(\ell)\varphi(\ell^{r_\ell}))^{-1} \sum_{u \bmod \ell^{r_\ell}} \chi_{0,\ell}(u) \prod_{i=1}^K \psi_{i,\ell}(F_i(u)).$$

Since $|\alpha_{\widehat{\chi}}| = \prod_{\ell^e \parallel q} |\alpha_{\widehat{\chi},\ell}|$, our claim (4.8) will follow once we show that for each prime power $\ell^e \parallel q$,

$$(4.10) \quad |\alpha_{\widehat{\chi},\ell}| \leq \begin{cases} (4D) \cdot \ell^{-r_\ell/D} = (4D) \cdot \ell^{-v_\ell(\mathfrak{f}(\widehat{\chi}))/D}, & \text{if } \ell > C_0, r_\ell \geq 1. \\ (2DC_0^{C_0+1}) \cdot \ell^{-r_\ell/D} = (2DC_0^{C_0+1}) \cdot \ell^{-v_\ell(\mathfrak{f}(\widehat{\chi}))/D}, & \text{if } 2 < \ell \leq C_0, r_\ell \geq \kappa + 1. \\ (DC_0^{C_0+1}) \cdot 2^{-r_2/D} = (DC_0^{C_0+1}) \cdot 2^{-v_2(\mathfrak{f}(\widehat{\chi}))/D} & \text{if } \ell = 2 \text{ divides } q, \text{ and } r_2 \geq \kappa + 1. \end{cases}$$

First consider $\ell > 2$. Let ψ_ℓ be a character generating the character group mod ℓ^{r_ℓ} . Then each $\psi_{i,\ell} = \psi_\ell^{A_i}$ for some $A_i \in [\varphi(\ell^{r_\ell})]$. Since at least one of $(\psi_{i,\ell})_{i=1}^K$ is primitive mod ℓ^{r_ℓ} , we must have

$$(4.11) \quad (A_1, \dots, A_K) \not\equiv \begin{cases} 0 \pmod{\ell-1}, & \text{if } r_\ell = 1 \\ 0 \pmod{\ell}, & \text{if } r_\ell \geq 2. \end{cases}$$

By (4.9), $\alpha_{\widehat{\chi}, \ell} = (\alpha(\ell)\varphi(\ell^{r_\ell}))^{-1} \sum_{u \bmod \ell^{r_\ell}} \psi_\ell(H(u))$ with $H(T) := T^{\varphi(\ell^{r_\ell})} \prod_{i=1}^K F_i(T)^{A_i}$. We bound the sum as follows: If $\ell > C_0$ and $r_\ell = 1$, we use Proposition 4.1, Lemma 4.3(1) and (4.11), noting that the prime divisors $\ell > C_0 > \max\{C, B_0\}$ satisfy $\gcd(\ell - 1, \beta) = 1$ by hypothesis $IFH(F_1, \dots, F_K; B_0)$; hence $|\sum_{u \bmod \ell} \psi_\ell(H(u))| \leq D\ell^{1/2} \leq D\ell^{1-1/D}$ for all $\ell > C_0$. If either $\ell > C_0, r_\ell \geq 2$ or $2 < \ell \leq C_0, r_\ell \geq \kappa + 1$, then Lemma 4.3(3) and (4.11) give both $\tau_\ell := \text{ord}_\ell(H') \leq \mathbb{1}_{\ell \leq C_0} C_0 \leq r_\ell - 2$ (as $\kappa > C_0 + 2$ by (4.4)) and $\sum_{\theta \in \mathbb{F}_\ell: H(\theta) \neq 0} \text{mult}_\theta(\ell^{-\tau_\ell} H') \leq D - 1$; hence $|\sum_{u \bmod \ell^{r_\ell}} \psi_\ell(H(u))| \leq D(C_0^{C_0})^{\mathbb{1}_{\ell \leq C_0}} \ell^{r_\ell(1-1/D)}$ by Proposition 4.2(i). Combining all this with (4.7) and $\varphi(\ell^{r_\ell}) \geq \ell^{r_\ell}/2$, we get the first two bounds in (4.10).

Consider the third case of (4.10), i.e. $2 \mid q$, $\alpha(2) = 1$, $\text{ord}_2(F) = 0$, $r_2 \geq \kappa + 1 > C_0 + 2D + 4$ by (4.7) and (4.4). Recall that the character group mod 2^{r_2} is generated by the characters ψ, η with $\psi(5) = \exp(2\pi i/2^{r_2-2})$, $\psi(-1) = 1$, $\eta(5) = 1$, $\eta(-1) = -1$. Thus each $\psi_{i,2} = \psi^{A_i} \eta^{B_i}$ with $A_i \in [2^{r_2-2}]$, $B_i \in [2]$; here $\mathbf{2} \nmid (\mathbf{A}_1, \dots, \mathbf{A}_K)$ as some $(\psi_{i,2})_{i=1}^K$ is primitive mod 2^{r_2} . By (4.9), $\alpha_{\widehat{\chi}, 2} = \varphi(2^{r_2})^{-1} \sum_{u \bmod 2^{r_2}} \psi \left(\prod_{i=1}^K F_i(u)^{A_i} \right) \eta \left(u^2 \prod_{i=1}^K F_i(u)^{B_i} \right)$. Writing $u = 4v + \lambda$ and using $\mathfrak{f}(\eta) = 4$ gives (4.12)

$$\alpha_{\widehat{\chi}, 2} = \frac{1}{2^{r_2-1}} \sum_{\substack{\lambda = \pm 1 \\ \prod_{i=1}^K F_i(\lambda) \equiv 1 \pmod{2}}} \eta \left(\prod_{i=1}^K F_i(\lambda)^{B_i} \right) \cdot \frac{1}{4} \sum_{v \bmod 2^{r_2}} \psi(g_\lambda(v)), \text{ with } g_\lambda(T) := \prod_{i=1}^K F_i(4T + \lambda)^{A_i}.$$

Now $g'_\lambda(T) = 4\widetilde{F}(4T + \lambda) \prod_{i=1}^K F_i(4T + \lambda)^{A_i-1}$ where $\widetilde{F} := \sum_{i=1}^K A_i F'_i \prod_{j \neq i} F_j$. For each λ in (4.12), $\prod_{i=1}^K F_i(4T + \lambda) \equiv \prod_{i=1}^K F_i(\lambda) \equiv 1 \pmod{2}$, so $t_\lambda := \text{ord}_2(g'_\lambda) = 2 + \text{ord}_2(\widetilde{F}(4T + \lambda))$. From the last two sentences, we get both $\text{mult}_\theta(2^{-t_\lambda} g'_\lambda) = \text{mult}_\theta(2^{-(t_\lambda-2)} \widetilde{F}(4T + \lambda)) \leq \deg(\widetilde{F}) \leq D - 1$ for any $\theta \in \mathbb{F}_2$, and

$$t_\lambda = 2 + \text{ord}_2(\widetilde{F}(4T + \lambda)) \leq 2D + \text{ord}_2(\widetilde{F}) \leq 2D + C_0 \leq \kappa - 3 \leq r_2 - 3;$$

here the first inequality uses the fact that $\text{ord}_2(W(4T + \lambda)) \leq \text{ord}_2(W) + 2\deg(W)$ for any $W \in \mathbb{Z}[T]$ (which can be seen by a straightforward divisibility argument), and the second inequality uses Lemma 4.3(2). Hence, using Proposition 4.2(ii) and the observations $t_\lambda \leq 2D + C_0$ and $\max_{\theta \in \mathbb{F}_2} \text{mult}_\theta(2^{-t_\lambda} g'_\lambda) \leq D - 1$ above, we can bound each inner sum in (4.12), thus obtaining the last bound in (4.10).

Proof of Proposition 4.4 (1). Since $\#\{\widehat{\chi} \bmod q : \mathfrak{f}(\widehat{\chi}) \mid m\} \leq m^K$, (4.8) yields for any $N \in \mathbb{Z}^+$,

$$(4.13) \quad \sum_{\substack{\widehat{\chi} \bmod q \\ \mathfrak{f}(\widehat{\chi}) \nmid Q_0}} |\alpha_{\widehat{\chi}}|^N \leq \sum_{\substack{m \mid q \\ m \nmid Q_0}} m^K \prod_{\substack{\ell \mid m \\ \ell > C_0}} \left(4D \cdot \ell^{-v_\ell(m)/D} \right)^N \prod_{\substack{\ell \leq C_0 \\ v_\ell(m) \geq \kappa+1}} \left(2DC_0^{C_0+1} \cdot \ell^{-v_\ell(m)/D} \right)^N.$$

Since Q_0 is the largest C_0 -smooth $(\kappa + 1)$ -free divisor of q , any m in (4.13) either: (i) has $P(m) > C_0$ or (ii) has $P(m) \leq C_0$ but is not $(\kappa + 1)$ -free. Let E_1 denote the contribution of all m of type (i) to the rightmost sum in (4.13), and E_2 be that of type (ii). Writing each m in E_1 as $m = pM$ for some $p > C_0$,

$$(4.14) \quad E_1 \leq \sum_{p > C_0} \frac{(4D)^N}{p^{N/D-K}} \sum_{M \mid q} M^K \prod_{\substack{\ell \mid M \\ \ell > C_0}} \left(4D \cdot \ell^{-v_\ell(M)/D} \right)^N \prod_{\substack{\ell \leq C_0 \\ v_\ell(M) \geq \kappa+1}} \left(2DC_0^{C_0+1} \cdot \ell^{-v_\ell(M)/D} \right)^N.$$

As $N \rightarrow \infty$, the first sum above is at most $(4D)^N \sum_{p > C_0} p^{-N/2D} \leq (4DC_0^{-1/4D})^N \sum_{p > C_0} p^{-5} \leq (4DC_0^{-1/4D})^N = o(1)$ by (4.4). Since $\prod_{\ell \leq C_0} (1 + \sum_{1 \leq v \leq \kappa} \ell^{vK}) \ll 1$, the second sum in (4.14) is

$$\begin{aligned} &\ll \prod_{\substack{\ell \mid q \\ \ell > C_0}} \left(1 + \sum_{1 \leq v \leq v_\ell(q)} \frac{(4D)^N}{\ell^{v(N/D-K)}} \right) \cdot \prod_{\substack{\ell \mid q \\ \ell \leq C_0}} \left(1 + \sum_{\kappa+1 \leq v \leq v_\ell(q)} \frac{(2DC_0^{C_0+1})^N}{\ell^{v(N/D-K)}} \right) \\ &\ll \exp \left((4DC_0^{-1/4D})^N \sum_{\ell > C_0} \sum_{v \geq 1} \ell^{-5v} \right) \cdot \exp \left((2DC_0^{C_0+1} 2^{-\kappa/4D})^N \sum_{\ell \leq C_0} \sum_{v \geq 1} \ell^{-5v} \right) \ll 1 \end{aligned}$$

by (4.4) and the fact that $N/D - K \geq N/(2D) \geq 10$. Hence overall $E_1 = o(1)$ as $N \rightarrow \infty$.

Any m counted in E_2 is C_0 -smooth but not $(\kappa + 1)$ -free, and hence $m = AB$ for some $(\kappa + 1)$ -full integer $A > 1$ and some $(\kappa + 1)$ -free integer B . Since $B \leq \prod_{\ell \leq C_0} \ell^\kappa \ll_{C_0, \kappa} 1$, and $N \rightarrow \infty$, we have

$$\begin{aligned} E_2 &\ll \sum_{\substack{A > 1: P(A) \leq C_0 \\ A \text{ is } (\kappa+1)\text{-full}}} \prod_{\ell \mid A} \frac{(2DC_0^{C_0+1})^N}{\ell^{v_\ell(A)(N/D-K)}} \leq \prod_{\ell \leq C_0} \left(1 + \sum_{v \geq \kappa+1} \frac{(2DC_0^{C_0+1})^N}{\ell^{Nv/2D}} \right) - 1 \\ &\leq \left| \exp \left(O \left(\sum_{\ell \leq C_0} \frac{(2DC_0^{C_0+1})^N}{\ell^{\kappa N/2D}} \right) \right) - 1 \right| \leq \left| \exp \left(O \left(\left(\frac{2DC_0^{C_0+1}}{2^{\kappa/2D}} \right)^N \right) \right) - 1 \right| \ll \left(\frac{2DC_0^{C_0+1}}{\kappa/2D} \right)^N, \end{aligned}$$

so that (4.4) yields $E_2 = o(1)$ as $N \rightarrow \infty$. Collecting all the estimates, we obtain subpart (1).

Proof of Proposition 4.4 (2). Fix any $r \in \mathbb{Z}^+$ and consider any $z \in (q, x]$. We may write

$$\sum_{\substack{P_1, \dots, P_r: P_1 \cdots P_r \leq x \\ P_1 > z, q < P_r < \cdots < P_1 \\ (\forall i) f_i(P_1) \cdots f_i(P_r) \equiv a_i \pmod{q}}} 1 \leq \sum_{(v_1, \dots, v_r) \in \mathcal{V}_{r,K}(q; \hat{a})} \sum_{\substack{q < P_r < \cdots < P_2 \leq x \\ (\forall j \geq 2) P_j \equiv v_j \pmod{q}}} \sum_{\substack{z < P_1 \leq x/P_2 \cdots P_r \\ P_1 \equiv v_1 \pmod{q}}} 1,$$

with $\mathcal{V}_{r,K}(q; \hat{a}) := \{(v_1, \dots, v_r) \in U_q^r : (\forall i) \prod_{j=1}^r F_i(v_j) \equiv a_i \pmod{q}\}$. By Brun–Titchmarsh, the innermost sum above is $\ll x/(\varphi(q)P_2 \cdots P_r \log(z/q))$; moreover, by Brun–Titchmarsh and partial summation, we have $\sum_{\substack{q < P_j \leq x \\ P_j \equiv v_j \pmod{q}}} 1/P_j \ll \log_2 x/\varphi(q)$ for each $j \in \{2, \dots, r\}$. Combining these two estimates,

$$(4.15) \quad \sum_{\substack{P_1, \dots, P_r: P_1 \cdots P_r \leq x \\ P_1 > z, q < P_r < \cdots < P_1 \\ (\forall i) f_i(P_1) \cdots f_i(P_r) \equiv a_i \pmod{q}}} 1 \ll \frac{\#\mathcal{V}_{r,K}(q; \hat{a})}{\varphi(q)^r} \cdot \frac{x(\log_2 x)^{r-1}}{\log(z/q)}.$$

Since $\#\mathcal{V}_{1,K}(q; \hat{a}) \ll q^{1-1/D_{\min}} \ll q^{-1/D_{\min}} \varphi(q) \log_2(3q)$ (by work of Konyagin [13, 14]⁶), we now get the last bound in (4.5). Next, detecting the congruences $\prod_{j=1}^r F_i(v_j) \equiv a_i \pmod{q}$ by orthogonality,

$$\begin{aligned} (4.16) \quad \varphi(q)^K \cdot \frac{\#\mathcal{V}_{r,K}(q; \hat{a})}{\varphi(q)^r} &= \alpha(q)^r \left| \sum_{\hat{\chi} \pmod{q}} \bar{\chi}_1(a_1) \cdots \bar{\chi}_K(a_K) (\alpha_{\hat{\chi}})^r \right| \leq \sum_{\hat{\chi} \pmod{q}} |\alpha_{\hat{\chi}}|^r \\ (4.17) \quad &\leq \sum_{M \mid q} M^K \prod_{\substack{\ell \mid M \\ \ell > C_0}} \left(4D \cdot \ell^{-v_\ell(M)/D} \right)^r \prod_{\substack{\ell \mid M: \ell \leq C_0 \\ v_\ell(M) \geq \kappa+1}} \left(2DC_0^{C_0+1} \cdot \ell^{-v_\ell(M)/D} \right)^r, \end{aligned}$$

where we used (4.8). The last sum above is precisely the second sum in (4.14), and hence again is

$$(4.18) \quad \ll \prod_{\substack{\ell \mid q \\ \ell \leq C_0}} \left(1 + \sum_{\kappa+1 \leq v \leq v_\ell(q)} \frac{(2DC_0^{C_0+1})^r}{\ell^{v(r/D-K)}} \right) \cdot \prod_{\substack{\ell \mid q \\ \ell > C_0}} \left(1 + \sum_{1 \leq v \leq v_\ell(q)} \frac{(4D)^r}{\ell^{v(r/D-K)}} \right).$$

If $r \geq KD + 1$, this is $\ll \exp(O(\sum_{\ell \mid q} \sum_{v \geq 1} \ell^{-v/D})) \leq \exp(O(\sum_{\ell \leq \omega(q)} \ell^{-1/D})) \leq \exp(O((\log q)^{1-1/D}))$. If $r \leq KD$, then (4.18) is $\ll \prod_{\ell \mid q} O(v_\ell(q) \ell^{v_\ell(q)(K-r/D)}) \leq q^{K-r/D} e^{O(\omega(q))} \prod_{\ell \mid q} v_\ell(q) \ll_\varepsilon q^{K-r/D+\varepsilon}$. Inserting these two bounds into (4.17), and then into (4.15), we get the first two bounds in (4.5).

⁶Konyagin's result can be restated as: For any fixed nonconstant $W \in \mathbb{Z}[T]$, we have $\#\{u \pmod{q} : W(u) \equiv b \pmod{q}\} \ll_W q^{1-1/\deg W}$ uniformly in q, b . We applied his result with W being the F_i with minimal degree $D_{\min}$.

Proof of Proposition 4.4 (3). A simpler version of the argument for (4.8) gives $|\alpha_{\widehat{\chi}}| \leq \prod_{\substack{\ell \nmid \widehat{\chi} \\ \ell > C_0}} (4D\ell^{-1/2})$ for any $\widehat{\chi} \bmod \text{squarefree } q$. With this, proceeding as above gives the first two bounds in (4.6). The third bound in (4.6) follows from (4.15) and $\#\mathcal{V}_{1,K}(q; \widehat{a}) = \prod_{\ell|q} \#\mathcal{V}_{1,K}(\ell; \widehat{a}) \ll (D_{\min})^{\omega(q)}$ (by the Chinese Remainder Theorem and the fact that $\#\{u \in \mathbb{F}_\ell : W(u) = 0\} \leq \deg(W)$ for all $W \in \mathbb{Z}[T]$ and $\ell \gg_W 1$).

It only remains to show the fourth bound in (4.6), i.e. for $r = 2$, $K = 1$, and $F_1 \in \mathbb{Z}[T]$ not squarefull. Proceeding as in the first paragraph of the proof of Lemma 4.3, we can find a constant C_1 exceeding the size of the leading coefficient of F_1 , such that F_1 has at least one simple root in $\overline{\mathbb{F}}_\ell$ for all $\ell > C_1$. We claim that $F_1(X)F_1(Y) - w$ is irreducible in $\overline{\mathbb{F}}_\ell[X, Y]$ for all $\ell > C_1$ and all $w \in U_\ell$. Indeed, suppose

$$(4.19) \quad F_1(X)F_1(Y) - w = U(X, Y)V(X, Y) = \left(\sum_{0 \leq j \leq r} u_j(Y)X^j \right) \left(\sum_{0 \leq j \leq s} v_j(Y)X^j \right) \text{ in } \overline{\mathbb{F}}_\ell[X, Y],$$

where $U(X, Y) = \sum_{j=0}^r u_j(Y)X^j$, $V(X, Y) = \sum_{j=0}^s v_j(Y)X^j \in \overline{\mathbb{F}}_\ell[X, Y]$. For any root $\theta \in \overline{\mathbb{F}}_\ell$ of F_1 , taking $Y = \theta$ in (4.19) shows that $U(X, \theta)$ and $V(X, \theta)$ are constant. Hence if both $r, s > 0$, then $u_r(\theta) = v_s(\theta) = 0$, giving $\prod_{\theta: F_1(\theta)=0} (Y - \theta) \mid \gcd(u_r(Y), v_s(Y))$; but then comparing the highest degree terms in

X on the two extremes of (4.19) yields $\prod_{\theta: F_1(\theta)=0} (Y - \theta)^2 \mid F_1(Y)$, contradicting that F_1 has a simple root

in $\overline{\mathbb{F}}_\ell$. Thus $rs = 0$. Assuming $s = 0$, we have $V(X, Y) = v_0(Y)$ with $F_1(X)F_1(Y) - w = U(X, Y)v_0(Y)$. Substituting a root of F_1 for X shows that $V(X, Y) = v_0(Y)$ is constant, proving the claimed irreducibility.

The Hasse–Weil bound in [17, Corollary 2b] now gives $\#\mathcal{V}_{2,1}(\ell; w) \leq \varphi(\ell)(1 + O(\ell^{-1/2}))$ uniformly in $\ell > C_1$, $w \in U_\ell$. Hence from $\#\mathcal{V}_{2,1}(q; a) = \prod_{\ell|q} \#\mathcal{V}_{2,1}(\ell; a)$ and (4.15), we get the last bound in (4.6). $\square$

4.1. Setting up for Theorems 1.2 and 1.3. The following estimates will be used repeatedly.

Lemma 4.5. [26, Proposition 2.1, Lemma 2.4] *Fix any multiplicative function $w : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ for which there exists a nonconstant $W \in \mathbb{Z}[T]$ satisfying $w(p) = W(p)$ for all primes p . Fix any $K_0 > 0$. Then for all $X \gg 1$ and uniformly in $q \leq (\log X)^{K_0}$ with $\alpha_W(q) := \varphi(q)^{-1} \#\{u \in U_q : W(u) \in U_q\} > 0$, we have*

$$\sum_{\substack{n \leq X \\ (w(n), q)=1}} 1 = \frac{X}{(\log X)^{1-\alpha_W(q)}} \exp(O((\log_2(3q))^{O(1)})), \quad \sum_{\substack{p \leq X \\ (W(p), q)=1}} \frac{1}{p} = \alpha_W(q) \log_2 X + O((\log_2(3q))^{O(1)}).$$

To show Theorems 1.2 and 1.3, we invoke Theorems 1.4 and 1.5 with $h_i = f_i$, $q \leq (\log x)^{K_0}$, $d = Q_0$, $y = \exp((\log x)^{\epsilon/2})$, $J = \lfloor \log_3 x \rfloor$, $z = x^{1/\log_2 x}$ (with Q_0 as after (4.4), K_0 and ϵ as in Theorems 1.2–1.3, and $\epsilon := 1$ for Theorem 1.3). Writing $\sum_{y < p \leq Y} \widehat{\chi}(\widehat{f}(p)) = \sum_{u \in U_q} \widehat{\chi}(\widehat{F}(u)) \sum_{\substack{y < p \leq Y \\ p \equiv u \pmod{q}}} 1$, Siegel–Walfisz gives (1.4) with $\alpha_{\widehat{\chi}}$ as in (4.3), $M = \varphi(q)$ and $\mathcal{E}(y) = \exp(-c\sqrt{\log y})$, with $c > 0$ an absolute constant.

Any quantity below is “negligible” if it is $o(\varphi(q)^{-K} \#\{n \leq x : (f(n), q) = 1\})$, with the number inside $o()$ estimated by Lemma 4.5. The right of (1.5) is negligible by Proposition 4.4(1). For $q \leq (\log x)^{K_0}$, the condition $P_J(n) > y$ in the *second* sum in (1.5) is ignorable upto negligible error, by (1.7) and the facts

$$(4.20) \quad \sum_{\substack{p \leq y \\ (f(p), q)=1}} \frac{1}{p} = \frac{\alpha(q)\epsilon}{2} \log_2 x + O((\log_2(3q))^{O(1)}); \quad \alpha(q) \gg \frac{1}{(\log_3 x)^D}; \quad \Psi(x, z) \leq \frac{x}{(\log x)^{(1+o(1)) \log_3 x}},$$

which use Lemma 4.5, (4.1), and known estimates on smooth numbers [3, p. 15], respectively. This gives

$$(4.21) \quad \sum_{\substack{n \leq x: P_J(n) > y \\ (\forall i) f_i(n) \equiv a_i \pmod{q}}} 1 - \left(\frac{\varphi(Q_0)}{\varphi(q)} \right)^K \sum_{\substack{n \leq x: (f(n), q) = 1 \\ (\forall i) f_i(n) \equiv a_i \pmod{Q_0}}} 1 = o \left(\frac{1}{\varphi(q)^K} \sum_{\substack{n \leq x \\ (f(n), q) = 1}} 1 \right).$$

For q in Theorem 1.2(i)-(ii), the condition $P_J(n) > y$ in (4.21) is ignorable upto negligible error by (1.6), (4.20), and the third bounds in (4.5), (4.6). For q, R in Theorem 1.3, the difference between the left of (1.2) and the first sum in (4.21) is the number $\#\{n \leq x : P_R(n) > q, P_J(n) \leq y, (\forall i) f_i(n) \equiv a_i \pmod{q}\}$ which is negligible by the last assertion of Theorem 1.5, (4.20), (4.5) and (4.6).⁷ The second asymptotic in (1.2) is clear, as the condition $P_R(n) \leq q$ forces $P_J(n) \leq y$ which can be dealt with by (1.7) and (4.20).

By (4.21) and the above observations, Theorems 1.2 and 1.3 would follow once we show that

$$(4.22) \quad \#\{n \leq x : (f(n), q) = 1, (\forall i) f_i(n) \equiv a_i \pmod{Q_0}\} \sim \varphi(Q_0)^{-K} \#\{n \leq x : (f(n), q) = 1\}.$$

This is clear if $Q_0 = 1$, or if $q^* := \prod_{\ell|q} \ell \ll 1$. (If $q^* \ll 1$, apply Theorem 1.1 to the integer $\text{lcm}[Q_0, q^*] \ll 1$). By orthogonality, (4.1) and Lemma 4.5, estimate (4.22) would follow once we show that: There exists a constant $\delta_0 := \delta_0(C_0, \kappa) < 1$ such that for all $q \leq (\log x)^{K_0}$ with $q \in \mathcal{Q}(\hat{f})$, $Q_0 > 1$ and $q^* \gg 1$,

$$(4.23) \quad \sum_{n \leq x} \mathbb{1}_{(f(n), q) = 1} \hat{\psi}(\hat{f}(n)) \ll \frac{x}{(\log x)^{1 - \delta_0 \alpha(q)}} \text{ for all nontrivial tuples of characters } \hat{\psi} \pmod{Q_0}.$$

If $\hat{\psi}(\hat{F}(u))$ is *not* constant on $\{u \in U_{Q_0} : F(u) \in U_{Q_0}\}$, then (4.23) follows by applying Proposition 1.6 to the character tuple $\hat{\chi} \pmod{q}$ induced by $\hat{\psi}$, and noting that $|\alpha_{\hat{\chi}}| = (\alpha(q)\varphi(q))^{-1} |\sum_{u \in U_q} \hat{\chi}(\hat{F}(u))| = (\alpha(Q_0)\varphi(Q_0))^{-1} |\sum_{u \in U_{Q_0}} \hat{\psi}(\hat{F}(u))|$ is bounded away from 1 by a *constant* (depending on C_0 and κ), since $Q_0 \ll_{C_0, \kappa} 1$. It thus only remains to show (4.23) when $\hat{\psi}(\hat{F}(u))$ is constant on $\{u \in U_{Q_0} : F(u) \in U_{Q_0}\}$.

5. MODIFYING THE LANDAU–SELBERG–DELANGE METHOD

We show (4.23) in the case when $\hat{\psi}(\hat{F}(u))$ takes a constant value $c_{\hat{\psi}}$ on $\{u \in U_{Q_0} : F(u) \in U_{Q_0}\}$. Recall that $q \leq (\log x)^{K_0}$ lies in $\mathcal{Q}(\hat{f})$, and has $q^* = \prod_{\ell|q} \ell \gg 1$ and $Q_0 > 1$. Note also that coprimality with q and q^* are equivalent, so that $\alpha(q^*) = \alpha(q) > 0$. We will modify the Landau–Selberg–Delange method, borrowing ideas from work of Scourfield [29]. We may assume throughout that $x \in \mathbb{Z}^+$.

5.1. Perron's formula and analytic continuations. We start by considering the Dirichlet series $\mathcal{F}(s) := \sum_{n \geq 1} \mathbb{1}_{(f(n), q) = 1} \hat{\psi}(\hat{f}(n)) / n^s$. By Perron's Formula [34, Theorem II.2.3], we have

$$(5.1) \quad \sum_{n \leq x} \mathbb{1}_{(f(n), q) = 1} \hat{\psi}(\hat{f}(n)) = \frac{1}{2\pi i} \int_{1 + \frac{1}{\log x} - iT}^{1 + \frac{1}{\log x} + iT} \frac{\mathcal{F}(s) x^s}{s} ds + O \left(x \sum_{n \geq 1} \frac{\mathbb{1}_{(f(n), q) = 1}}{n^{1 + \frac{1}{\log x}} (1 + T |\log(x/n)|)} \right)$$

uniformly in $T \geq 1$. The total contribution of all $n \notin (3x/4, 5x/4)$ to the O -term above is

$$(5.2) \quad \ll \frac{x}{T} \sum_{n \geq 1} \frac{1}{n^{1 + 1/\log x}} \ll \frac{x}{T} \exp \left(\sum_p \frac{1}{p^{1 + 1/\log x}} \right) \ll \frac{x}{T} \exp \left(\sum_{p \leq x} \frac{1}{p} + \sum_{j \geq 0} \exp(-2^j) \sum_{x^{2^j} < p \leq x^{2^{j+1}}} \frac{1}{p} \right),$$

which is $\ll x(\log x)/T$. Next, for any $n \in (3x/4, x - 1]$, we have $n = x - v$ for some $v \in [1, x/4] \cap \mathbb{Z}$, so that $|\log(x/n)| = -\log(1 - v/x) \gg v/x$. Thus, the contribution of all such n to the O -term in (5.1) is

⁷Here we need the inequalities $\max\{s + r/D, (KD + 1) - s - r + r/D\} \geq K + 1/2$ for $r \in [KD]$, and $\max\{s + r/2, (2K + 1) - s - r/2\} \geq K + 1/2$, which can both be checked easily.

$\ll T^{-1} \sum_{1 \leq v < x/4} x/v \ll x(\log x)/T$; likewise, so is the contribution of all $n \in [x+1, 5x/4]$. Hence

$$(5.3) \quad \sum_{n \leq x} \mathbb{1}_{(f(n), q)=1} \widehat{\psi}(\widehat{f}(n)) = \frac{1}{2\pi i} \int_{1+\frac{1}{\log x}-iT}^{1+\frac{1}{\log x}+iT} \frac{\mathcal{F}(s)x^s}{s} ds + O\left(\frac{x \log x}{T}\right).$$

For $s \in \mathbb{C}$, write $\sigma := \operatorname{Re}(s)$ and $t := \operatorname{Im}(s)$. Define $\mathcal{L}_m(t) := \log(m(|t|+1))$. Let $c_1 > 0$ be an absolute constant such that any $m \in \mathbb{Z}^+$, the product $\prod_{\eta \bmod m} L(s, \eta)$ has at most one zero ρ_e (the ‘‘Siegel zero’’) in the region $\sigma > 1 - c_1/\mathcal{L}_m(t)$; here $\rho_e \in \mathbb{R}$ is simple and is associated to a real character $\eta_e \bmod m$. In what follows, any term(s) involving ρ_e or η_e can be ignored if ρ_e does not exist. Now if $(\alpha(q^*), c_{\widehat{\psi}}) \neq (1, 1)$, then we cut the complex plane along the ray $(-\infty, 1]$. If $(\alpha(q^*), c_{\widehat{\psi}}) = (1, 1)$ and ρ_e exists, then we cut the complex plane along the ray $(-\infty, \rho_e]$. In the remaining case, we make no branch cut.

Lemma 5.1. *For $\sigma > 1$, we have $\mathcal{F}(s) = F_{\widehat{\psi}}(s)G_{\widehat{\psi}}(s)$, where $F_{\widehat{\psi}}(s) := \prod_{\eta \bmod q^*} L(s, \eta)^{\alpha(q^*)c_{\widehat{\psi}}\gamma(\eta)}$, $\gamma(\eta) := (\alpha(q^*)\varphi(q^*))^{-1} \sum_{u: uF(u) \in U_{q^*}} \bar{\eta}(u)$, and $G_{\widehat{\psi}}$ is analytic on $\sigma > 1 - 1/\log q^*$ with $G_{\widehat{\psi}}(s) \ll (\log_2 q^*)^3$ uniformly therein. Hence $\mathcal{F}(s)$ continues analytically into the region $\{s : \sigma > 1 - c_1/\mathcal{L}_{q^*}(t)\} - [1 - c_1, 1]$.*

Proof. By orthogonality mod q^* and the identity $\log L(s, \eta) = \sum_{p, v \geq 1} \eta(p^v)/vp^{vs}$ for $\sigma > 1$, we have

$$(5.4) \quad \sum_{p \equiv b \pmod{q^*}} \frac{1}{p^s} = \frac{1}{\varphi(q^*)} \sum_{\eta \bmod q^*} \bar{\eta}(b) \log L(s, \eta) - \sum_{\substack{p, v \geq 2 \\ p^v \equiv b \pmod{q^*}}} \frac{1}{vp^{vs}}.$$

Moreover, by the Euler product of the Dirichlet series $\mathcal{F}(s)$, we have for $\sigma > 1$,

$$(5.5) \quad \mathcal{F}(s) = \prod_{p: pF(p) \in U_{q^*}} \left(1 - \frac{c_{\widehat{\psi}}}{p^s}\right)^{-1} \cdot \prod_p \left(1 + \sum_{v \geq 1} \frac{\mathbb{1}_{(f(p^v), q)=1} \widehat{\psi}(\widehat{f}(p^v))}{p^{vs}}\right) \left(1 - \frac{c_{\widehat{\psi}} \mathbb{1}_{pF(p) \in U_{q^*}}}{p^s}\right).$$

Expanding the logarithm of the first product into Taylor series, and splitting the sum $\sum_{p: pF(p) \in U_{q^*}} p^{-s}$ as $\sum_{b \in U_{q^*}: F(b) \in U_{q^*}} \sum_{p \equiv b \pmod{q^*}} p^{-s}$, we see by (5.4) that this logarithm equals

$$(5.6) \quad \alpha(q^*)c_{\widehat{\psi}} \sum_{\eta \bmod q^*} \gamma(\eta) \log L(s, \eta) + \left(\sum_{\substack{p, v \geq 2 \\ pF(p) \in U_{q^*}}} \frac{(c_{\widehat{\psi}})^v}{vp^{vs}} - \sum_{\substack{p, v \geq 2 \\ pF(p^v) \in U_{q^*}}} \frac{c_{\widehat{\psi}}}{vp^{vs}} \right).$$

Hence $\mathcal{F}(s) = F_{\widehat{\psi}}(s)G_{\widehat{\psi}}(s)$, where $G_{\widehat{\psi}}(s)$ is the second product in (5.5) times the exponential of the parenthesized expression in (5.6). Now since $\sum_{p, v \geq 2} p^{-2v/3} \ll 1$, the last two sums in (5.6) are analytic functions of size $O(1)$ on $\sigma > 2/3$. The contribution of all $p \nmid q^*$ to the second product in (5.5) is exactly $\prod_{p \nmid q^*} (1 + \sum_{v \geq 2} p^{-vs} \{\mathbb{1}_{(f(p^v), q^*)=1} \widehat{\psi}(\widehat{f}(p^v)) - c_{\widehat{\psi}} \mathbb{1}_{(F(p)f(p^{v-1}), q^*)=1} \cdot \widehat{\psi}(\widehat{f}(p^{v-1}))\})$), which is analytic of size $O(1)$ on $\sigma > 2/3$. Finally for $\sigma > 1 - 1/\log q^*$, the total contribution of all $p \mid q^*$ to the second product in (5.5) has size $\leq \exp(\sum_{p \mid q^*} \sum_{v \geq 1} p^{-vs}) \ll \exp(\sum_{p \mid q^*} p^{-1} \exp(\log p / \log q^*)) \ll \exp(e \sum_{p \leq p_{\omega(q^*)}} p^{-1}) \ll (\log_2 q^*)^3$ with $p_{\omega(q^*)}$ being the $\omega(q^*)$ -th prime. Collecting all these observations completes the proof. $\square$

Lemma 5.2. *Fix $W \in \mathbb{Z}[T]$ nonconstant. Consider squarefree $Q > 1$ with $\alpha_W(Q) := \varphi(Q)^{-1} \#\{u : uW(u) \in U_Q\} > 0$. Let $\gamma(\eta) = (\alpha_W(Q)\varphi(Q))^{-1} \sum_{u: uW(u) \in U_Q} \bar{\eta}(u)$. For s with $\sigma > 1 - c_1/2\mathcal{L}_Q(t)$, we have $|\sum_{\eta \bmod Q} \alpha_W(Q)\gamma(\eta)L'(s, \eta)/L(s, \eta) + \alpha_W(Q)/(s-1) - \alpha_W(Q)\gamma(\eta_e)/(s-\rho_e)| \ll \mathcal{L}_Q(t)$.*

This follows from Scourfield [29, Lemmas 15(i), 9(ii)], taking ‘‘ ξ ’’ there to be $e^{6\mathcal{L}_Q(t)}$, and using $L(z, \eta) = L(z, \eta') \prod_{\ell \mid Q} (1 - \eta'(\ell)/\ell^z)$ and $\gamma(\eta) = \gamma(\eta')$ when $\eta \bmod Q$ is induced by the primitive character η' . Note that if $\alpha_W(Q) = 1$ (which happens if, say, $\alpha_W(Q) > 0$ and $W(T) = c \cdot T^k$ for some constants c, k), then $\{u : uW(u) \in U_Q\} = U_Q$, so that $\gamma(\eta) = 0$ for all nontrivial $\eta \bmod Q$, and Lemma 5.2 is a classical result.

In our setting, we consider the function $H_{\hat{\psi}}(s) = F_{\hat{\psi}}(s) (s-1)^{\alpha(q^*)c_{\hat{\psi}}} (s-\rho_e)^{-\alpha(q^*)\gamma(\eta_e)c_{\hat{\psi}}}$ which analytically continues into $\sigma > 1 - c_1/\mathcal{L}_{q^*}(t)$. Lemma 5.2 yields $|H'_{\hat{\psi}}(z)/H_{\hat{\psi}}(z)| \ll \mathcal{L}_{q^*}(\text{Im}(z))$ uniformly in $z \in \mathbb{C}$ with $\text{Re}(z) > 1 - c_1/2\mathcal{L}_{q^*}(\text{Im}(z))$. Since $\log |H_{\hat{\psi}}(w)/H_{\hat{\psi}}(s)| \leq \left| \int_{\text{Re}(s)}^{\text{Re}(w)} |H'_{\hat{\psi}}(u+it)/H_{\hat{\psi}}(u+it)| du \right|$, we get

$$(5.7) \quad H_{\hat{\psi}}(s) \asymp H_{\hat{\psi}}(w), \text{ uniformly in } s, w \in \mathbb{C} \text{ having } \text{Im}(s) = \text{Im}(w) =: t$$

$$\text{and } 1 - c_1/2\mathcal{L}_{q^*}(t) \leq \text{Re}(s) \leq \text{Re}(w) \leq 1 + 100/\mathcal{L}_{q^*}(t).$$

5.2. The contour shift. We assume that $\rho_e \bmod q^*$ exists and $(\alpha(q^*), c_{\hat{\psi}}) \neq (1, 1)$, otherwise much simpler versions of our argument suffice. (We mention the modifications for the other possibilities at the end of this section.) By scaling c_1 if necessary, we may then *also* assume that $\rho_e > 1 - 5c_1/(24 \log q^*)$, and that $f(\eta_e)$ (which is squarefree) is large enough that it is divisible by a prime exceeding $D + 2$.

Define $T = \exp(\sqrt{\log x})$, $\tilde{\rho}_e = (2 + \rho_e)/3$, and $\vartheta(t) = 1 - c_1/(4\mathcal{L}_{q^*}(t))$. Let $\delta_e, \delta > 0$ be any parameters satisfying $\vartheta(0) < \rho_e - 2\delta_e$ and $\rho_e + 2\delta_e < \tilde{\rho}_e < 1 - 2\delta$. (We will eventually let $\delta, \delta_e \rightarrow 0+$.) Consider the contour Γ_1 consisting of the segment $\Gamma_2 = \{\sigma + iT : \sigma \in [\vartheta(T), 1 + 1/\log x]\}$, the curve $\Gamma_3 = \{\vartheta(t) + it : t \in [0, T]\}$, the segments $\Gamma_4 = [\vartheta(0), \rho_e - \delta_e]$, $\Gamma_6 = [\rho_e + \delta_e, \tilde{\rho}_e]$ and $\Gamma_7 = [\tilde{\rho}_e, 1 - \delta]$ all *above* the branch cut, the semicircle $\Gamma_5 = \{\rho_e + \delta_e \cdot e^{i\theta} : 0 < \theta < \pi\}$, the circle $\Gamma_8 = \{1 + \delta \cdot e^{i\theta} : -\pi < \theta < \pi\}$, and the reflections $\bar{\Gamma}_j$ of Γ_j about the real line (for $2 \leq j \leq 7$), with all components directed as in Figure 1.

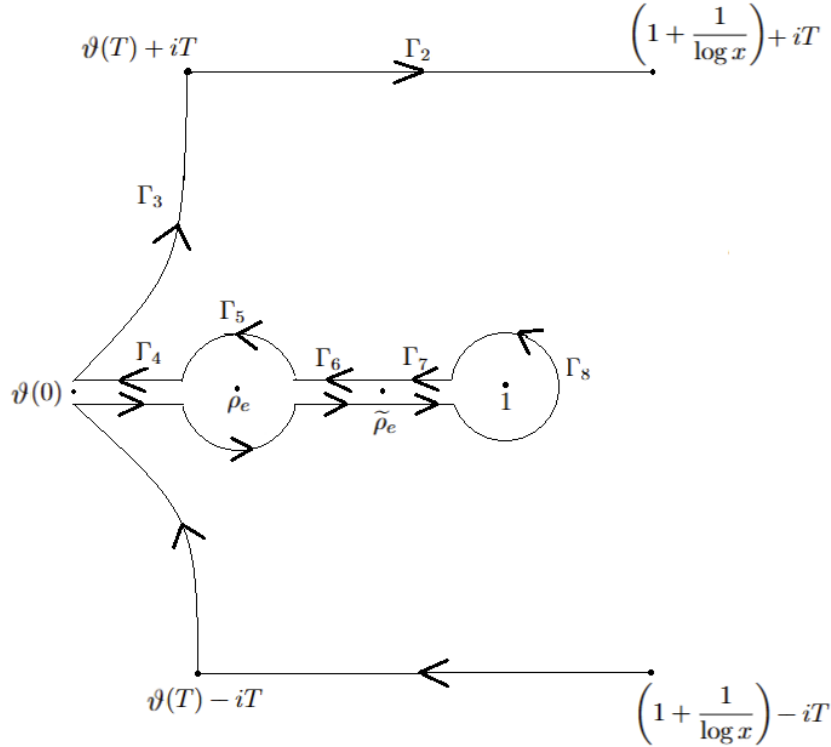


FIGURE 1. The Contour Γ_1 in the case when the Siegel-zero ρ_e exists and $(\alpha(q^*), c_{\hat{\psi}}) \neq (1, 1)$

By Cauchy's integral theorem and the final assertion of Lemma 5.1, we now obtain from (5.3),

$$(5.8) \quad \sum_{n \leq x} \mathbb{1}_{(f(n), q)=1} \hat{\psi}(\hat{f}(n)) = \frac{1}{2\pi i} \int_{\Gamma_1} \frac{\mathcal{F}(s)x^s}{s} ds + O\left(\frac{x \log x}{T}\right).$$

We will show that all parts of Γ_1 except Γ_7 and $\bar{\Gamma}_7$ are negligible. The following bounds will be useful.

Proposition 5.3. (1) We have $|\mathcal{F}(s)| \ll \log x$ uniformly in $s \in \Gamma_2 \cup \Gamma_3 \cup \bar{\Gamma}_2 \cup \bar{\Gamma}_3$.
(2) For any fixed $\epsilon > 0$, we have $|H_{\hat{\psi}}(s)| \ll_{\epsilon} (\log x)^{\epsilon \alpha(q^*)/5}$ uniformly in real $s \in [1 - c_1/(4 \log q^*), 1]$.

Proof. (1) Let $\mu(t) := 1 + c_1/(7\mathcal{L}_{q^*}(t))$. Consider any $s \in \Gamma_2 \cup \Gamma_3 \cup \bar{\Gamma}_2 \cup \bar{\Gamma}_3$. By Lemma 5.1, we have $\mathcal{F}(s) \ll (\log_2 q^*)^3 |F_{\hat{\psi}}(s)| \ll (\log_2 q^*)^3 |H_{\hat{\psi}}(s)| \cdot |s - 1|^{-\alpha(q^*)\text{Re}(c_{\hat{\psi}})} |s - \rho_e|^{\alpha(q^*)\text{Re}(\gamma(\eta_e)c_{\hat{\psi}})}$. Hence by (5.7),

$$(5.9) \quad \mathcal{F}(s) \ll (\log_2 q^*)^3 \cdot |H_{\hat{\psi}}(\mu(t) + it)| \cdot |s - 1|^{-\alpha(q^*)\text{Re}(c_{\hat{\psi}})} |s - \rho_e|^{\alpha(q^*)\text{Re}(\gamma(\eta_e)c_{\hat{\psi}})}.$$

Using the Dirichlet series of $\log L(\mu(t) + it, \eta)$ and interchanging sums, we see that $|F_{\hat{\psi}}(\mu(t) + it)| = |\exp(\sum_{\eta \bmod q^*} \gamma(\eta) \alpha(q^*) c_{\hat{\psi}} \log L(\mu(t) + it, \eta))| \leq \exp(\sum_{p, r \geq 1} r^{-1} p^{-r\mu(t)} |\sum_{\eta \bmod q^*} \alpha(q^*) \gamma(\eta) \eta(p^r)|)$. Now

$$\sum_{\eta \bmod q^*} \alpha(q^*) \gamma(\eta) \eta(p^r) = \frac{1}{\varphi(q^*)} \sum_{u: uF(u) \in U_{q^*}} \sum_{\eta \bmod q^*} \bar{\eta}(u) \eta(p^r) = \sum_{u: uF(u) \in U_{q^*}} \mathbb{1}_{p^r \equiv u \bmod q^*} = \mathbb{1}_{p^r F(p^r) \in U_{q^*}}.$$

Hence $|F_{\hat{\psi}}(\mu(t) + it)| \leq \exp(\sum_{p, r \geq 1} r^{-1} p^{-r\mu(t)}) = \zeta(\mu(t)) = 1/(\mu(t) - 1) + O(1) \ll \mathcal{L}_{q^*}(t)$. By the definition of $H_{\hat{\psi}}(\mu(t) + it)$, we thus obtain the following bound, which holds uniformly in all $t \in \mathbb{R}$.

$$(5.10) \quad |H_{\hat{\psi}}(\mu(t) + it)| \ll \mathcal{L}_{q^*}(t) |\mu(t) + it - 1|^{\alpha(q^*)\text{Re}(c_{\hat{\psi}})} |\mu(t) + it - \rho_e|^{-\alpha(q^*)\text{Re}(\gamma(\eta_e)c_{\hat{\psi}})}.$$

Finally, for $\theta \in \{1, \rho_e\}$ and $s = \sigma + it \in \Gamma_2 \cup \Gamma_3 \cup \bar{\Gamma}_2 \cup \bar{\Gamma}_3$, we have $|s - \theta| \asymp |\mu(t) + it - \theta|$. (This is clear for $|t| \geq 1/100$. If $|t| < 1/100$, then $s \in \Gamma_3 \cup \bar{\Gamma}_3$, and this follows from $\rho_e > 1 - 5c_1/(24 \log q^*)$.) Inserting the last estimate and (5.10) into (5.9), we get $\mathcal{F}(s) \ll (\log_2 q^*)^3 \mathcal{L}_{q^*}(t) \ll (\log_3 x)^3 \log(q^*T) \ll \log x$.

(2) By (5.7) and (5.10), $|H_{\hat{\psi}}(s)| \ll |H_{\hat{\psi}}(\mu(0))| \ll (\log q^*)^3 (1 - \rho_e)^{-\alpha(q^*)}$ for $s \in [1 - c_1/(4 \log q^*), 1]$. The desired bound now follows from $1 - \rho_e \gg (q^*)^{-\epsilon/20K_0} \gg (\log x)^{-\epsilon/20}$ (by Siegel's Theorem) and (4.1). $\square$

Proposition 5.4. Set $I_j := \int_{\Gamma_j} \mathcal{F}(s) x^s / s \, ds$ and $\bar{I}_j := \int_{\bar{\Gamma}_j} \mathcal{F}(s) x^s / s \, ds$. Then

$$(5.11) \quad |I_2| + |I_3| + |\bar{I}_2| + |\bar{I}_3| + |I_4 + \bar{I}_4| + |I_6 + \bar{I}_6| \ll x \exp(-(\log x)^{1/3}).$$

$$\lim_{\delta_e \rightarrow 0+} I_5 = \lim_{\delta_e \rightarrow 0+} \bar{I}_5 = \lim_{\delta \rightarrow 0+} I_8 = 0.$$

Proof. The bounds on $I_2, I_3, \bar{I}_2, \bar{I}_3$ are immediate by Proposition 5.3(1) and $|s| \gg t + 1$ for $s \in \Gamma_3 \cup \bar{\Gamma}_3$. To prove (5.11), we only show the bound for $I_4 + \bar{I}_4$; the argument for $I_6 + \bar{I}_6$ is analogous. For $s \in \Gamma_4$, we have $(s - 1)^{-\alpha(q^*)c_{\hat{\psi}}} \cdot (s - \rho_e)^{\alpha(q^*)\gamma(\eta_e)c_{\hat{\psi}}} = |1 - s|^{-\alpha(q^*)c_{\hat{\psi}}} \cdot |\rho_e - s|^{\alpha(q^*)\gamma(\eta_e)c_{\hat{\psi}}} \cdot e^{-i\pi\alpha(q^*)c_{\hat{\psi}} + i\pi\alpha(q^*)\gamma(\eta_e)c_{\hat{\psi}}}$; the same holds for $s \in \bar{\Gamma}_4$ with the opposite sign of the power of e . Hence by the definition of $H_{\hat{\psi}}(s)$,

$$(5.12) \quad |I_4 + \bar{I}_4| \ll \int_{\vartheta(0)}^{\rho_e - \delta_e} \frac{|H_{\hat{\psi}}(u) G_{\hat{\psi}}(u)| x^u}{u} (1 - u)^{-\alpha(q^*)\text{Re}(c_{\hat{\psi}})} \cdot (\rho_e - u)^{\alpha(q^*)\text{Re}(\gamma(\eta_e)c_{\hat{\psi}})} \, du$$

$$(5.13) \quad \ll x^{\rho_e} (\log x)^{\alpha(q^*)/200} \cdot (\log_2 q^*)^3 \cdot (1 - \rho_e)^{-\alpha(q^*)} \int_{\vartheta(0)}^{\rho_e - \delta_e} (\rho_e - u)^{\alpha(q^*)\text{Re}(\gamma(\eta_e)c_{\hat{\psi}})} \, du$$

by Lemma 5.1 and Proposition 5.3(2). By our hypothesis in the first paragraph of § 5.2, the conductor $f(\eta_e)$ has a prime factor $\ell_0 > D + 2$. As such, factoring $\eta_e = \prod_{\ell|q^*} \eta_{e,\ell}$ with $\eta_{e,\ell}$ a character mod ℓ ,

$$(5.14) \quad |\alpha(q^*)\gamma(\eta_e)| = \prod_{\ell|q^*} \frac{1}{\varphi(\ell)} \left| \sum_{\substack{u \in U_{\ell} \\ F(u) \in U_{\ell}}} \eta_{e,\ell}(u) \right| \leq \frac{1}{\ell_0 - 1} \left| - \sum_{\substack{u \in U_{\ell_0} \\ F(u) \equiv 0 \pmod{\ell_0}}} \eta_{e,\ell_0}(u) \right| \leq \frac{D}{D + 1}.$$

Hence the integral in (5.13) is at most $(\rho_e - \vartheta(0))^{1 + \alpha(q^*)\text{Re}(\gamma(\eta_e)c_{\hat{\psi}})} / (1 + \alpha(q^*)\text{Re}(\gamma(\eta_e)c_{\hat{\psi}})) \ll 1$. The bound on $|I_4 + \bar{I}_4|$ in (5.11) now follows from $1 - \rho_e \gg (\log x)^{-1/200}$ (Siegel's Theorem) and (4.1).

To show $\lim_{\delta_e \rightarrow 0+} I_5 = 0$, let $M_{\hat{\psi}}$ be the maximum of $|H_{\hat{\psi}}(s)|$ on a closed disk centered at ρ_e that contains Γ_5 and is contained in the region $\sigma > 1 - c_1/4\mathcal{L}_{q^*}(t)$. Then $M_{\hat{\psi}}$ is finite as $H_{\hat{\psi}}(s)$ is holomorphic on the region $\sigma > 1 - c_1/\mathcal{L}_{q^*}(t)$. Parametrizing Γ_5 as $s = \rho_e + \delta_e \cdot e^{i\theta}$ for $0 < \theta < \pi$, invoking Lemma 5.1 and arguing as above, we find that $|I_5| \ll M_{\hat{\psi}} (\log_2 q^*)^3 x^{\rho_e + \delta_e} \cdot (1 - \rho_e - \delta_e)^{-\alpha(q^*)} \cdot (\delta_e)^{1 + \alpha(q^*)\operatorname{Re}(\gamma(\eta_e)c_{\hat{\psi}})}$. Hence (5.14) yields $\lim_{\delta_e \rightarrow 0+} I_5 = 0$. The arguments for $\bar{I}_5$ and I_8 are analogous: For I_8 , we only need the fact that $\lim_{\delta \rightarrow 0+} \delta^{1 - \alpha(q^*)\operatorname{Re}(c_{\hat{\psi}})} = 0$, since $(\alpha(q^*), c_{\hat{\psi}}) \neq (1, 1)$ as assumed in the first paragraph of § 5.2. $\square$

Letting $\delta, \delta_e \rightarrow 0+$ and invoking Proposition 5.4, we thus obtain from (5.8),

$$(5.15) \quad \sum_{n \leq x} \mathbb{1}_{(f(n), q)=1} \hat{\psi}(\hat{f}(n)) = \frac{1}{2\pi i} \int_{\tilde{\Gamma}} \frac{\mathcal{F}(s)x^s}{s} ds + O\left(x \exp(-(\log x)^{1/3})\right),$$

with $\tilde{\Gamma}$ being the union of the two horizontal segments joining the points 1 and $\tilde{\rho}_e$ above and below the branch cut, directed like $\Gamma_7 \cup \bar{\Gamma}_7$. (To see that the integral in (5.15) converges, we start as in (5.12), use Lemma 5.1 and Proposition 5.3(2), and note that $\int_{\tilde{\rho}_e}^1 (1-u)^{\operatorname{Re}(\alpha(q^*)c_{\hat{\psi}})} du$ is finite as $(\alpha(q^*), c_{\hat{\psi}}) \neq (1, 1)$.)

5.3. Bounding the integral in (5.15). Starting analogously to (5.12), we have for any fixed $\epsilon > 0$,

$$(5.16) \quad \left| \int_{\tilde{\Gamma}} \frac{\mathcal{F}(s)x^s}{s} ds \right| \ll \int_{\tilde{\rho}_e}^1 \frac{|H_{\hat{\psi}}(u)G_{\hat{\psi}}(u)|x^u}{u} (1-u)^{-\alpha(q^*)\operatorname{Re}(c_{\hat{\psi}})} \cdot (u - \rho_e)^{\alpha(q^*)\operatorname{Re}(\gamma(\eta_e)c_{\hat{\psi}})} du \\ \ll (\log x)^{\epsilon\alpha(q^*)/4} \int_{\tilde{\rho}_e}^1 |G_{\hat{\psi}}(u)| \cdot x^u (1-u)^{-\alpha(q^*)\operatorname{Re}(c_{\hat{\psi}})} du,$$

where we used Proposition 5.3(2) and $(u - \rho_e)^{\alpha(q^*)\operatorname{Re}(\gamma(\eta_e)c_{\hat{\psi}})} \ll (1 - \rho_e)^{-\alpha(q^*)} \ll_{\epsilon} (\log x)^{\epsilon\alpha(q^*)/20}$.

Set $\epsilon_0 := \max_{1 < r \leq C_0^{C_0\kappa}} \cos(2\pi/r) < 1$. Since $Q_0 \in (1, C_0^{C_0\kappa}]$ and $c_{\hat{\psi}}$ is a $\varphi(Q_0)$ -th root of unity, we must have $\operatorname{Re}(c_{\hat{\psi}}) \leq \epsilon_0$ or $c_{\hat{\psi}} = 1$. If $\operatorname{Re}(c_{\hat{\psi}}) \leq \epsilon_0$, using Lemma 5.1 and writing $u = 1 - v/\log x$ yields from (5.16),

$$(5.17) \quad \left| \int_{\tilde{\Gamma}} \frac{\mathcal{F}(s)x^s}{s} ds \right| \ll \frac{x(\log_3 x)^3}{(\log x)^{1 - \alpha(q^*)(\epsilon_0 + \epsilon/4)}} \cdot \Gamma(1 - \alpha(q^*)\operatorname{Re}(c_{\hat{\psi}})) \ll \frac{x}{(\log x)^{1 - \alpha(q^*)(\epsilon_0 + \epsilon/3)}}$$

as the Γ -function is $O(1)$ on the interval $[1 - \epsilon_0, 1]$. Inserting (5.17) into (5.15) proves (4.23) when $c_{\hat{\psi}} \neq 1$.

If $c_{\hat{\psi}} = 1$, then $\hat{\psi}(\hat{F}(u)) = 1$ on $\{u \in U_{Q_0} : F(u) \in U_{Q_0}\}$. Applying the definition of $\mathcal{Q}(\hat{f})$ to the characters mod q induced by $\hat{\psi} = (\psi_1, \dots, \psi_K)$, we get $\mathbb{1}_{(f(2^j), q)=1} \hat{\psi}(\hat{f}(2^j)) = -1$ for all $j \geq 1$. Isolating $p = 2$ from the second product in (5.5), we thus have $G_{\hat{\psi}}(s) = G_{\hat{\psi},1}(s)G_{\hat{\psi},2}(s)$, with $G_{\hat{\psi},2}(s) := (1 + \sum_{j \geq 1} \mathbb{1}_{(f(2^j), q)=1} \hat{\psi}(\hat{f}(2^j))/2^{js}) (1 - c_{\hat{\psi}} \mathbb{1}_{(2F(2), q^*)=1}/2^s)$ analytic on $\sigma > 0$ satisfying $G_{\hat{\psi},2}(1) = 0$, and with $G_{\hat{\psi},1}(s)$ analytic on $\sigma > 1 - 1/\log q^*$ satisfying $|G_{\hat{\psi},1}(s)| \ll (\log_2 q^*)^3$ uniformly therein. (The assertions on $G_{\hat{\psi},1}$ can be shown by following the arguments in Lemma 5.1.) Hence for all $u \in [\tilde{\rho}_e, 1]$,

$$(5.18) \quad |G_{\hat{\psi}}(u)| \ll (\log_2 q^*)^3 \cdot |G_{\hat{\psi},2}(1) - G_{\hat{\psi},2}(u)| \ll (\log_3 x)^3 \int_u^1 |G'_{\hat{\psi},2}(w)| dw \ll (\log_3 x)^3 (1 - u).$$

Inserting this bound into (5.16), and then making the substitution $u = 1 - v/\log x$, we obtain

$$(5.19) \quad \left| \int_{\tilde{\Gamma}} \frac{\mathcal{F}(s)x^s}{s} ds \right| \ll \frac{x(\log_3 x)^3}{(\log x)^{2 - \alpha(q^*)(1 + \epsilon/4)}} \cdot \Gamma(2 - \alpha(q^*)) \ll \frac{x}{(\log x)^{2 - \alpha(q^*)(1 + \epsilon/3)}},$$

which by (5.15) completes the proof of (4.23) in the remaining case $c_{\hat{\psi}} = 1$.

Finally, if $(\alpha(q^*), c_{\hat{\psi}}) = (1, 1)$ and ρ_e exists, then our branch cut is only along $(-\infty, \rho_e]$, so we replace all Γ_j and $\bar{\Gamma}_j$ (for $j \geq 5$) by a circle of radius δ_e centered at ρ_e , going from below to above the branch cut; we

eventually let $\delta_e \rightarrow 0+$. If ρ_e doesn't exist, then for $(\alpha(q^*), c_{\widehat{\psi}}) \neq (1, 1)$ we replace Γ_5 and $\overline{\Gamma}_5$ by straight segments above and below the branch cut (which is along $(-\infty, 1]$), while for $(\alpha(q^*), c_{\widehat{\psi}}) = (1, 1)$, we take $\Gamma_1 = \Gamma_2 + \Gamma_3 + \overline{\Gamma}_2 + \overline{\Gamma}_3$ (there is no branch cut in this case). This establishes Theorems 1.2 and 1.3. $\square$

6. OPTIMALITIES IN THEOREMS 1.2 AND 1.3

Lemma 6.1. *Let $w_1, \dots, w_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ be multiplicative and $\{W_i\}_i \subset \mathbb{Z}[T]$ be nonconstant such that $W := \prod_{i=1}^K W_i$ is squarefree and $w_i(p) = W_i(p)$ for all i and all primes p . There exists a constant $C_{\widehat{W}}$ depending only on $\widehat{W} := (W_1, \dots, W_K)$ such that any $q \in \mathbb{Z}^+$ with $P^-(q) > C_{\widehat{W}}$ lies in $\mathcal{Q}(\widehat{w})$.*

Proof. If $P^-(q) > 1 + \sum_i \deg W_i$, then $\varphi(q)^{-1} \#\{u : uW(u) \in U_q\} \geq \prod_{\ell|q} (1 - (\sum_i \deg W_i)/(\ell - 1)) > 0$, so $\mathcal{T}(q) := \{(W_1(u), \dots, W_K(u)) : uW(u) \in U_q\} \neq \emptyset$. Now $\mathcal{T}(q)$ maps to $\prod_{\ell^e \parallel q} \mathcal{T}(\ell^e)$ under the canonical map $U_q^K \cong \prod_{\ell^e \parallel q} U_{\ell^e}^K$, so if $\mathcal{T}(q)$ doesn't generate U_q^K , then by [22, Lemma 5.13], there exists $\ell^e \parallel q$ and a nontrivial character tuple $\widehat{\psi} = (\psi_1, \dots, \psi_K) \pmod{\ell^e}$ with $\widehat{\psi}(\widehat{W}(u)) = \prod_{i=1}^K \psi_i(W_i(u))$ constant on $\{u : uW(u) \in U_{\ell^e}\}$. By [20, Lemma 5], this constancy is impossible if $\ell \gg_{\widehat{W}} 1$ (independently of e). Hence $P^-(q) \gg_{\widehat{W}} 1$ ensures that $\mathcal{T}(q)$ generates U_q^K , forcing $q \in \mathcal{Q}(\widehat{w})$ vacuously. $\square$

As in Theorems 1.2 and 1.3, consider any multiplicative $f_1, \dots, f_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ and any nonconstant $\{F_i\}_{i=1}^K \subset \mathbb{Z}[T]$ satisfying $f_i(p) = F_i(p)$ for all i and all (primes) p . Fix any $\epsilon > 0$. We say that a quantity “grows too fast” if it grows strictly faster than the expected main term $\varphi(q)^{-K} \#\{n \leq x : (\prod_{i=1}^K f_i(n), q) = 1\}$ as $x \rightarrow \infty$, in the given ranges of q . Note that for any F_i , the functions $(f_1, \dots, f_K)$ are not WUD modulo any $q > (\log x)^{(1+\epsilon)\alpha(q)/(K-1)}$: Indeed, fix any $b \in \mathbb{Z} \setminus \{0\}$ with $\prod_{i=1}^K F_i(b) \neq 0$. Consider any q with $P^-(q) > \max\{C_{\widehat{F}}, 1 + |b \prod_{i=1}^K F_i(b)|\}$, with $C_{\widehat{F}}$ as in Lemma 6.1. Then $q \in \mathcal{Q}(\widehat{f})$. But by Siegel–Walfisz, $\#\{n \leq x : (\forall i) f_i(n) \equiv F_i(b) \pmod{q}\} \geq \#\{p \leq x : p \equiv b \pmod{q}\} \gg x/(\varphi(q) \log x)$ which grows too fast if $q > (\log x)^{(1+\epsilon)\alpha(q)/(K-1)}$. (To see this, use Lemma 4.5, (4.1) and $\varphi(q) \gg q/\log_2 q$.)

Below, we construct nonconstant $\{F_i\}_{i=1}^K \subset \mathbb{Z}[T]$ with $\prod_{i=1}^K F_i$ separable (so multiplicative independence and IFH hypotheses are trivial). Fix any $r \in \mathbb{Z}_{>1}$, $K_0 \gg_{F_1, \dots, F_K} 1$, and assume $q \leq (\log x)^{K_0}$.

Range optimality in Theorem 1.2(i). Let $F_i(T) := (T-1)^r + i$ (so $D_{\min} = r$), and let q be any r -th power with $P^-(q) > \max\{C_{\widehat{F}}, 2K\}$. Then $\#\{n \leq x : (\forall i) f_i(n) \equiv i \pmod{q}\} \geq \#\{p \leq x : p \equiv 1 \pmod{q^{1/r}}\} \gg x/(q^{1/r} \log x)$ which, by Lemma 4.5 and (4.1), grows too fast if $q > (\log x)^{(1+\epsilon)\alpha(q)(K-1/r)^{-1}}$.

Range optimality in Theorem 1.2(ii). Let $F_i(T) := \prod_{j=1}^r (T-2j) + 2(2i-1)$ which is irreducible (by Eisenstein's criterion at 2). Consider any squarefree q with $P^-(q) > \max\{C_{\widehat{F}}, 4Kr\}$. By the Chinese Remainder Theorem, $\prod_{j=1}^r (v-2j) \equiv 0 \pmod{q}$ has exactly $r^{\omega(q)}$ distinct solutions, all in U_q . Hence

$$(6.1) \quad \sum_{\substack{n \leq x \\ (\forall i) f_i(n) \equiv 2(2i-1) \pmod{q}}} 1 \geq \sum_{\substack{n \leq x: P(n) > q \\ (\forall i) f_i(n) \equiv 2(2i-1) \pmod{q}}} 1 \geq \sum_{\substack{q < P \leq x \\ \prod_{j=1}^r (P-2j) \equiv 0 \pmod{q}}} 1 \gg \frac{r^{\omega(q)} x}{\varphi(q) \log x}$$

which by Lemma 4.5 and (4.1), grows too fast if $q^{K-1} D_{\min}^{\omega(q)} = q^{K-1} r^{\omega(q)} > (\log x)^{(1+\epsilon)\alpha(q)}$.

For completeness and later use, we construct squarefree $q \rightarrow \infty$ satisfying the stronger conditions $P^-(q) > \max\{C_{\widehat{F}}, 4^r Kr!\}$ and $r^{\omega(q)} > (\log x)^{2\alpha(q)}$. Since $4^r Kr! > |F_1(0)|$, the Prime Ideal Theorem gives

$$\prod_{\max\{C_{\widehat{F}}, 4^r Kr!\} < \ell \leq Z} \alpha(\ell) \leq \prod_{\max\{C_{\widehat{F}}, 4^r Kr!\} < \ell \leq Z} \left(1 - \frac{1}{\ell-1} \sum_{\substack{u \pmod{\ell} \\ F_1(u) \equiv 0 \pmod{\ell}}} 1\right) \leq \frac{b_r}{\log Z} \text{ for all } Z \gg 1,$$

for a constant $b_r > 0$. Fix $K_0 > 16b_r$. Let $Y := (K_0/2) \log_2 x$; $q := \prod_{\max\{C_{\widehat{F}}, 4^r K r!\} < \ell \leq Y} \ell \leq (\log x)^{K_0}$. Then $\omega(q) \geq Y/(2 \log Y) > 4b_r \log_2 x / \log Y$ and $\alpha(q) \leq b_r / \log Y$. Hence $r^{\omega(q)} > e^{\omega(q)/2} > (\log x)^{2\alpha(q)}$.

Optimality in Theorem 1.3. By (6.1), the $R = 2$ in (1.3) can't be reduced to 1. We show that $R = 2K + 1$ in (1.3) can't be reduced to $2K - 1$. With $F_i(T) = \prod_{j=1}^r (T - 2j) + 2(2i - 1)$ as above, consider multiplicative $f_1, \dots, f_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ with $f_i(p) = F_i(p)$ and $f_i(p^2) = 1$ for all i, p . Construct q as above. Let $n = (p_1 \cdots p_{K-1})^2 P \leq x$, where p_i, P are primes with $q < p_{K-1} < \cdots < p_1 < x^{1/4K} < x^{1/3} < P$ and $\prod_{j=1}^r (P - 2j) \equiv 0 \pmod{q}$. Then $P_{2K-1}(n) > q$ and $f_i(n) \equiv 2(2i - 1) \pmod{q}$, whence by Siegel–Walfisz,

$$(6.2) \quad \sum_{\substack{n \leq x: P_{2K-1}(n) > q \\ (\forall i) f_i(n) \equiv 2(2i-1) \pmod{q}}} 1 \gg \frac{r^{\omega(q)} x}{\varphi(q) \log x} \frac{1}{(K-1)!} \sum_{\substack{p_1, \dots, p_{K-1} \in (q, x^{1/4K}) \\ p_1, \dots, p_{K-1} \text{ distinct}}} \frac{1}{(p_1 \cdots p_{K-1})^2}.$$

The sum of $1/(p_1 \cdots p_{K-1})^2$ over all $p_1, \dots, p_{K-1} \in (q, x^{1/4K})$ is $(\sum_{q < p \leq x^{1/4K}} p^{-2})^{K-1} \gg (q \log q)^{-(K-1)}$, and over all $p_1, \dots, p_{K-1} > q$ with p_i not all distinct is $\ll (\sum_{p > q} p^{-4}) \cdot (\sum_{p > q} p^{-2})^{K-3} \ll q^{-K}$. Hence

$$(6.3) \quad \sum_{\substack{n \leq x: P_{2K-1}(n) > q \\ (\forall i) f_i(n) \equiv 2(2i-1) \pmod{q}}} 1 \gg \frac{r^{\omega(q)}}{\varphi(q)(q \log q)^{K-1}} \cdot \frac{x}{\log x} \gg \frac{r^{\omega(q)}}{\varphi(q)^K} \cdot \frac{x}{(\log_2 x)^K \log x}$$

as $\varphi(q) \gg q / \log_2 q$. Since $r^{\omega(q)} > (\log x)^{2\alpha(q)}$, the last expression grows too fast.

7. THE GENERAL RESULTS

The hypothesis $\alpha(q) \neq 0$ in Theorems 1.1-1.3 prevented the input sets from becoming too sparse (Lemma 4.5) and guaranteed that the behavior of the f_i at the *primes* was most significant. If $\alpha(q) = 0$, then the behavior of the f_i at a *higher prime power* becomes more significant. Narkiewicz's general criterion in [20] allows for this flexibility, and we obtain optimal uniform analogues of his general criterion as well.

7.1. Extending Theorems 1.4 and 1.5 for certain sparse input sets. Fix $\nu \in \mathbb{Z}^+$ and $C > 0$. Given $h_1, \dots, h_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ and $q \in \mathbb{Z}^+$, we say that $\widehat{h} := (h_1, \dots, h_K)$ is ν -full (with respect to q) if the ν -free part of any n satisfying $(h_1(n) \cdots h_K(n), q) = 1$ is at most C . Equivalently, any such $n = n_1 n_2$, where $n_1 \leq C$ and n_2 is a ν -full integer. We will just write “ $\widehat{h}$ are ν -full”; the “ q ” will always be clear from the context. In this section, the control of the h_i at the ν -th powers of primes is most significant: All hypotheses, results and arguments are modified accordingly. In particular, taking $\nu = 1$ here recovers everything before. Note that if $\widehat{h}$ is ν -full, then $\widehat{h}$ is also $(\nu - 1)$ -full; in particular, any $\widehat{h}$ is 1-full.

In what follows, let $\Psi_\nu(x, z)$ denote the number of ν -full, z -smooth integers up to x ; we bound this in Lemma 7.2. Given any ν -full integer-valued multiplicative functions $(h_1, \dots, h_K)$, and any $q, r \in \mathbb{Z}^+$ and $z > 0$, we redefine $\xi_r(q; z) := \xi_r(q; z; \{h_i\}_{i=1}^K)$ be *any* parameter satisfying, for all $x \geq z$ and all $a_i \in U_q$,

$$(7.1) \quad \#\{P_1 \cdots P_r \leq x : P_i \text{ primes, } P_1 > z, q < P_r < \cdots < P_1, (\forall i) h_i(P_1^\nu) \cdots h_i(P_r^\nu) \equiv a_i \pmod{q}\} \\ \leq \xi_r(q; z) \cdot x(\log \log x)^r / \log(z/q).$$

We retain the notations $\widehat{\chi} = (\chi_1, \dots, \chi_K)$, $\mathfrak{f}(\widehat{\chi}) = \text{lcm}[\mathfrak{f}(\chi_1), \dots, \mathfrak{f}(\chi_K)]$, and $\widehat{\chi}(\widehat{h}(n)) = \prod_{i=1}^K \chi_i(h_i(n))$.

Theorem 7.1. *Consider any $q \in \mathbb{Z}^+$ with $q > C$. Consider any ν -full multiplicative $\widehat{h} := (h_1, \dots, h_K)$, any $M \geq 1$, $y \geq q$, and a decreasing function $\mathcal{E} : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, such that for all $\widehat{\chi} \bmod q$ and $Y \geq y$,*

$$(7.2) \quad \sum_{y \leq p \leq Y} \widehat{\chi}(\widehat{h}(p^\nu)) = \alpha_{\widehat{\chi}} \sum_{y \leq p \leq Y} \chi_0(h(p^\nu)) + O(MY\mathcal{E}(y)), \text{ with } \alpha_{\widehat{\chi}} \text{ in the unit disk.}$$

Then (1.5) holds as stated, only $P_J(n) > y$, Jx/y , $x\mathcal{E}(y)$ replaced by $P_{J\nu}(n) > y$, $J(x/y)^{1/\nu}$, $x^{1/\nu}\mathcal{E}(y)$.

The assertions of Theorem 1.5 also hold as stated, only by assuming that $\hat{h}$ is ν -full, and replacing $P_J(n) \leq y$, $x(2 \log \log x)^\cdots$, $h(p)$, $\Psi(x, z)$, x/y , $\xi_R(q; z) + R^K \sum_{\substack{1 \leq r < R \\ 1 \leq s < K}} \xi_r(q; z)/q^{\max\{s, R-r-s\}}$ by $P_{J\nu}(n) \leq y$, $x^{1/\nu}(2 \log \log x)^\cdots$, $h(p^\nu)$, $\Psi_\nu(x, z)$, $(x/y)^{1/\nu}$, $\xi_{\lceil R/\nu \rceil}(q; z) + R^K \sum_{\substack{1 \leq r < R/\nu \\ 1 \leq s < K\nu}} \xi_r(q; z)/q^{\max\{s/\nu, R/\nu-r-s\}}$.

All implied constants depend only on ν, K, C and the implied constant in (7.2).

Proof. The proof is analogous; we just mention the nontrivial changes. We say that $n \leq x$ is “convenient” (“conv”) if $n = m(P_J \cdots P_1)^\nu$ where (2.1) holds; note that this representation is unique. Any inconvenient n with $P_{J\nu}(n) > y$ and $(h(n), q) = 1$ must be divisible by the $(\nu + 1)$ -th power of a prime exceeding y (since the ν -free part of n is at most $C < y$). Now by the Erdős–Székere estimate [9] on ν -full numbers,

$$(7.3) \quad \sum_{\substack{n \leq x: (h(n), q)=1 \\ \exists p > y \text{ s.t. } p^{\nu+1} | n}} 1 \leq \sum_{\substack{p > y \\ r \geq \nu+1}} \sum_{\substack{m \leq x/p^r \\ (h(m), q)=1}} 1 \leq \sum_{\substack{p > y \\ r \geq \nu+1}} \sum_{m_1 \leq C} \sum_{\substack{m_2 \leq x/p^r \\ m_2 \text{ is } \nu\text{-full}}} 1 \ll x^{1/\nu} \sum_{\substack{p > y \\ r \geq \nu+1}} \frac{1}{p^{r/\nu}} \ll \left(\frac{x}{y}\right)^{1/\nu}.$$

Hence, to prove the analogue of (1.5), we may replace all “ $P_{J\nu}(n) > y$ ” by “ n conv”. Now (2.2) and (2.3) hold as stated, while (2.4) and (2.5) hold with all x/m , $\hat{h}(P_i)$, $h(P_i)$ replaced by $x^{1/\nu}/m^{1/\nu}$, $\hat{h}(P_i^\nu)$, $h(P_i^\nu)$. Hence, the analogue of the expression “ $x/y + J^{-1}Mx\mathcal{E}(y) \log x$ ” mentioned after (2.5) is

$$\sum_{n \leq x/y^\nu: (h(n), q)=1} 1 + J^{-1}Mx^{1/\nu}\mathcal{E}(y) \sum_{n \leq x/y^\nu: (h(n), q)=1} n^{-1/\nu}.$$

The first of the two sums above is $\ll \sum_{n_1 \leq C} \sum_{\substack{n_2 \leq x/y^\nu \\ n_2 \text{ is } \nu\text{-full}}} 1 \ll x^{1/\nu}/y$. The second of the two sums is

$$(7.4) \ll \sum_{\substack{n \leq x \\ (h(n), q)=1}} \frac{1}{n^{1/\nu}} \leq \sum_{n_1 \leq C} \frac{1}{n_1^{1/\nu}} \sum_{\substack{n_2 \leq x \\ n_2 \text{ is } \nu\text{-full}}} \frac{1}{n_2^{1/\nu}} \ll \prod_{p \leq x} \left(1 + \sum_{r \geq \nu} \frac{1}{p^{r/\nu}}\right) \ll \exp\left(\sum_{p \leq x} \frac{1}{p}\right) \ll \log x.$$

Proceeding as in the rest of the proof of Theorem 1.4 yields the analogue of (2.6), and hence of (1.5).

We generalize the last assertion of Theorem 1.5 (for any $R \leq J$). Since $\#\{n \leq x : P(n) \leq z, (h(n), q) = 1\} \leq \sum_{n_1 \leq C} \sum_{\substack{n_2 \leq x: P(n_2) \leq z \\ n_2 \text{ is } \nu \text{ full}}} 1 \ll \Psi_\nu(x, z)$, we restrict to n with $P(n) > z$ and with a $(\nu + 1)$ -free y -rough part (by (7.3)). Redefine $\sum_{n \leq x}^*$ only by replacing “squarefree y -rough part” and “ $P_J(n) \leq y$ ” by “ $(\nu + 1)$ -free y -rough part” and “ $P_{J\nu}(n) \leq y$ ”. By ν -fullness of $\hat{h}$, the y -rough part of any n counted in $\sum_{n \leq x}^*$ must then be of the form A^ν with A squarefree and $P^-(A) > y$. Next, $E_{r,s}$ is redefined and (2.7) holds with “ $p \parallel n$ ”, “ $p^2 \mid n$ ”, “ R ”, “ K ” replaced by $p^\nu \parallel n$, $p^{\nu+1} \mid n$, R/ν , $K\nu$. (Note that the q -rough part of n is ν -full, by ν -fullness of $\hat{h}$.) Write any n counted in $E_{r,s}$ as $mp_1^{c_1} \cdots p_s^{c_s} A^\nu$, only with “ $c_j \geq 2$ ”, “ $h_i(A)$ ”, “ $\sum_{j=1}^s c_j \geq R - r$ ” replaced by $c_j \geq \nu + 1$, $h_i(A^\nu)$, $\sum_{j=1}^s c_j \geq R - \nu r$. Finally, to show the analogues of (2.8) and (2.9), we write $m = m_1 m_2$ with $m_1 \leq C$ and with m_2 being ν -full, and note that

$$\sum_{\substack{m_2 \leq x: P_{J\nu}(m_2) \leq y \\ m_2 \text{ is } \nu\text{-full} \\ p > y \Rightarrow p^{\nu+1} \nmid m_2}} \frac{\mathbb{1}_{(h(m_2), q)=1}}{m_2^{1/\nu}} \leq \sum_{\substack{A \leq x \\ \Omega(A) \leq J}} \frac{1}{A} \sum_{\substack{B: P(B) \leq y \\ B \text{ is } \nu\text{-full}}} \frac{\mathbb{1}_{(h(B), q)=1}}{B^{1/\nu}} \ll (2 \log_2 x)^J \exp\left(\sum_{p \leq y} \frac{\mathbb{1}_{(h(p^\nu), q)=1}}{p}\right),$$

again since the y -rough part of m_2 must be of the form A^ν with $P^-(A) > y$ and $\Omega(A) \leq J$. $\square$

We now bound $\Psi_\nu(x, z)$, generalizing some known bounds on smooth numbers (such as [3, p. 15]).

Lemma 7.2. *Fix $\nu \in \mathbb{Z}^+$. As $x, z \rightarrow \infty$, we have $\Psi_\nu(x, z) \ll x^{1/\nu}(\log z) \exp(-u\nu^{-1} \log u + O(u \log_2(3u)))$, uniformly for $(\log x)^{\max\{3, 2\nu\}} \leq z \leq x^{1/2}$. Here $u := \log x / \log z$.*

Proof. Let $\eta := \log u / \log z \leq \min\{1/3, 1/(2\nu)\}$. Since $\sum_p \sum_{r \geq \nu+1} p^{-r(1-\eta)/\nu} \ll \sum_p p^{-(1-\eta)(1+1/\nu)} \ll 1$,

$$\Psi_\nu(x, z) \leq \sum_{\substack{n: P(n) \leq z \\ n \text{ is } \nu\text{-full}}} \left(\frac{x}{n}\right)^{(1-\eta)/\nu} \leq x^{(1-\eta)/\nu} \prod_{p \leq z} \left(1 + \sum_{r \geq \nu} \frac{1}{p^{r(1-\eta)/\nu}}\right) \ll x^{(1-\eta)/\nu} \exp\left(\sum_{p \leq z} \frac{1}{p^{1-\eta}}\right).$$

Now $\sum_{p \leq z} p^{-(1-\eta)} = \log_2 z + \sum_{p \leq z} (\exp(\eta \log p) - 1)/p + O(1)$. In the last sum, the contribution of all $p \leq 2^{1/\eta}$ is $\ll \eta \sum_{p \leq 2^{1/\eta}} \log p / p \ll 1$, while that of $p \in (2^{1/\eta}, z]$ is at most $(\exp(\eta \log z) - 1) \sum_{2^{1/\eta} < p \leq z} 1/p \leq u(\log_2(3u) + O(1))$. Collecting all these bounds completes the proof. $\square$

Proposition 7.3. *Assuming (7.2) and the sentences preceding it, we have for any $\hat{\chi} \bmod q$ and $x \geq y$,*

$$(7.6) \quad \sum_{n \leq x} \hat{\chi}(\hat{h}(n)) \ll \frac{x^{1/\nu}}{\log x} \exp\left(\sum_{\substack{p \leq y \\ (h(p^\nu), q)=1}} \frac{1}{p} + |\alpha_{\hat{\chi}}| \sum_{\substack{y < p \leq x \\ (h(p^\nu), q)=1}} \frac{1}{p} + O(M(\log x)^3 \mathcal{E}(y))\right) \\ + \left(\frac{x}{y}\right)^{1/\nu} + x^{1/\nu} (\log x)^2 \exp(-u(2\nu)^{-1} \log u + O(u \log_2(3u))),$$

where $u = \log x / \log y$. The implied constant depends only on ν, C and the implied constant in (1.4).

Proof. Again by (7.3) and ν -fullness of $\hat{h}$, we may restrict to $n \leq x$ whose y -rough part is of the form A^ν for some squarefree A . Any such n is uniquely written as $n_1 B A^\nu$, with $n_1 \leq C$ being ν -free, B being ν -full, A squarefree, and $P(B) \leq y < P^-(A)$. The total contribution of those n for which $B > x^{1/2}$ is at most $\sum_{n_1 \leq C} \sum_{A \leq (\sqrt{x}/n_1)^{1/\nu}} \Psi_\nu(x/(n_1 A^\nu), y)$, which by Lemma 7.2 is absorbed in the right of (7.6), as $x/(n_1 A^\nu) \geq x^{1/2}$. On the other hand, the total contribution of those n which have $B \leq x^{1/2}$ is exactly

$$\sum_{\substack{n_1 \leq C \\ n_1 \text{ is } \nu\text{-free}}} \hat{\chi}(\hat{h}(n_1)) \sum_{\substack{B \leq x^{1/2}: P(B) \leq y \\ B \text{ is } \nu\text{-full}; (h(B), q)=1}} \hat{\chi}(\hat{h}(B)) \sum_{A \leq (x/n_1 B)^{1/\nu}} \mu(A)^2 \mathbb{1}_{P^-(A) > y} \hat{\chi}(\hat{h}(A^\nu)).$$

Arguing as in section 3 to bound the innermost sum, and bounding $\sum 1/B^{1/\nu}$ as in (7.5), we are done. $\square$

7.2. Narkiewicz's general criterion and uniform analogues. Consider multiplicative $f_1, \dots, f_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$, and $\{F_{i,j}\}_{1 \leq i \leq K, 1 \leq j \leq V} \subset \mathbb{Z}[T]$ satisfying $f_i(p^j) = F_{i,j}(p)$ for all $i \in [K]$, $j \in [V]$ and primes p . Let

$\alpha_j(q) := \varphi(q)^{-1} \#\{u \in U_q : \prod_{i=1}^K F_{i,j}(u) \in U_q\}$. For any fixed $\nu \in [V]$, we say that q is ν -admissible if $\alpha_\nu(q) \neq 0$ but $\alpha_j(q) = 0$ for all $j < \nu$. (i.e. ν is the minimal index j for which $\prod_{i=1}^K F_{i,j}$ carries some unit to some unit). Let $\mathcal{Q}(\nu; \hat{f})$ be the set of all ν -admissible q satisfying the following property:

$$(7.7) \quad \text{For all } \hat{\chi} \neq (\chi_0, \dots, \chi_0) \bmod q \text{ satisfying } \prod_{i=1}^K \chi_i(F_{i,\nu}(u)) = 1 \text{ on } \{u \in U_q : \prod_{i=1}^K F_{i,\nu}(u) \in U_q\}, \\ \text{there exists a prime } p \text{ such that } \sum_{m \geq 0} \hat{\chi}(\hat{f}(p^m)) p^{-m/\nu} = 0.$$

By the triangle inequality, such $p \leq 2^\nu$; hence $\mathcal{Q}(1; \hat{f}) = \mathcal{Q}(\hat{f})$, and $(F_{i,\nu})_{i=1}^K$ play the role of $(F_i)_{i=1}^K$ from sections 1-6. For $K = 1$, we use $\mathcal{Q}(\nu; f_1)$ to denote $\mathcal{Q}(\nu; \hat{f})$. We now state Narkiewicz's general criterion.

Theorem 7.4. [20] *Fix ν -admissible q . Then $(f_1, \dots, f_K)$ are WUD mod q if and only if $q \in \mathcal{Q}(\nu; \hat{f})$.*

Our first uniform analogue of Theorem 7.4 is as follows; all implied constants depend only on $\{F_{i,j}\}_{1 \leq i \leq K, 1 \leq j \leq \nu}$.

Theorem 7.5. *Theorem 1.2 holds, with F_i , $\mathcal{Q}(\hat{f})$, $\alpha(q)$ replaced by $F_{i,\nu}$, $\mathcal{Q}(\nu; \hat{f})$, $\alpha_\nu(q)$ respectively.*

All claims on optimality and ranges of q following the statement of Theorem 1.2 hold for *any* fixed ν , with analogous changes. The examples are also analogous to those in § 6; we discuss the changes in § 7.5.

Theorem 7.6. *Fix $K_0, B_0 > 0$. Assume $(F_{i,\nu})_{i=1}^K$ are multiplicatively independent. Then (1.2) holds uniformly in $q \leq (\log x)^{K_0}$ with $q \in \mathcal{Q}(\nu; \hat{f})$ satisfying $IFH(F_{1,\nu}, \dots, F_{K,\nu}; B_0)$, and uniformly in $a_i \in U_q$.*

$$(7.8) \quad \text{Here } R = \begin{cases} \nu(KD + 1), & \text{if } \nu < D := \sum_{i=1}^K \deg F_{i,\nu}. \\ \text{the least integer exceeding } \nu(1 + (\nu + 1)(K - 1/D)), & \text{if } \nu \geq D. \end{cases}$$

If q is squarefree, then we can take $R = \nu(K\nu + K - \nu + 1) + 1$, and we can further improve this to

$$(7.9) \quad R = \begin{cases} 2, & \text{if } K = \nu = 1 \text{ and } F_{1,\nu} \text{ is not squarefull.} \\ \nu(K\nu + K - \nu) + 1, & \text{if } \nu > 1 \text{ and at least one of } \{F_{i,\nu}\}_{i=1}^K \text{ is not squarefull.} \end{cases}$$

Both values of R in (7.9) are optimal in general; see § 7.5. The *second* value of R in (7.8) is also optimal, as shown by the $\sigma(n)$ example below. Note that the second values of R in (7.8) and (7.9) had no analogues in Theorem 1.3. We now mention concrete examples which can be addressed in this greater generality.

1. For $\sigma(n)$, Śliwa [33] gets $\mathcal{Q}(1; \sigma) = \{q : 2 \nmid q\}$; $\mathcal{Q}(2; \sigma) = \{q : 2 \mid q, 3 \nmid q\}$; $\mathcal{Q}(\nu; \sigma) = \emptyset$ for $\nu > 2$. Theorem 1.2 dealt with odd q . For $q \in \mathcal{Q}(2; \sigma)$, Theorem 7.5 shows that σ is WUD mod $q \leq (\log x)^{(2-\delta)\tilde{\alpha}(q)}$, and mod squarefree $q \leq (\log x)^{K_0}$ with $2^{\omega(q)} \leq (\log x)^{(1-\delta)\tilde{\alpha}(q)}$, where $\tilde{\alpha}(q) = \prod_{\ell \equiv 1 \pmod{3}} \ell | q (\ell - 3)/(\ell - 1)$.

The latter range is optimal [30, §7.2]. By Theorem 7.6, the ranges extend to $q \leq (\log x)^{K_0}$ if we restrict to n with $P_5(n) > q$ ($P_3(n) > q$ for squarefree q); both these restrictions are optimal [30, §6.1, §7.1].

2. For $\sigma_3(n) = \sum_{d|n} d^3$, Narkiewicz [21] gets $\mathcal{Q}(1; \sigma_3) = \{q : (q, 14) = 1\}$; $\mathcal{Q}(2; \sigma_3) = \{q : (q, 6) = 2\}$; $\mathcal{Q}(\nu; \sigma_3) = \emptyset$ for all $\nu > 2$. For $q \in \mathcal{Q}(1; \sigma_3)$, our results show that σ_3 is WUD mod $q \leq (\log x)^{(3/2-\delta)\tilde{\alpha}(q)}$, and mod squarefree $q \leq (\log x)^{K_0}$ with $3^{\omega(q)} \leq (\log x)^{(1-\delta)\tilde{\alpha}(q)}$, where $\tilde{\alpha}(q) := \prod_{\ell \equiv 1 \pmod{3}} \ell | q (\ell - 4)/(\ell - 1)$.

$\cdot \prod_{\substack{\ell | q \\ \ell \not\equiv 1 \pmod{3}}} (\ell - 2)/(\ell - 1)$. The ranges extend to $q \leq (\log x)^{K_0}$ if we restrict to n with $P_4(n) > q$ ($P_2(n) > q$ for squarefree q). Analogous consequences can be given for $q \in \mathcal{Q}(2; \sigma_3)$ varying with x .

Our results apply to the functions $\sigma_r(n)$, and to families like $(n, \varphi(n), \sigma(n))$ ($\sigma, \sigma_2, \sigma_3$), $(\varphi, \sigma, \sigma_2, \sigma_3)$, etc. In general, Theorems 7.5 and 7.6 convert explicit (fixed-modulus) descriptions of $\mathcal{Q}(\nu; \hat{f})$ into Siegel–Walfisz analogues. However, such descriptions remain unknown in general. In fact, even just for σ_r (with $r > 1$ fixed), the sets $\{\mathcal{Q}(\nu; \sigma_r)\}_{\nu=1}^\infty$ have only been computed upto $r \leq 200$ by Narkiewicz and Rayner [21, 27, 28]. Narkiewicz [19, 22] shows that for $(f_1, \dots, f_K)$ satisfying some natural additional constraints, we have $\mathcal{Q}(\nu; \hat{f}) = \emptyset$ for all $\nu \gg 1$; he also gives algorithms to compute the nonempty $\mathcal{Q}(\nu; \hat{f})$.

7.3. Additional ingredients for Theorems 7.5 and 7.6. To use the results in § 7.1, note that:

$$(7.10) \quad \text{If } q \text{ is } \nu\text{-admissible, then } (f_1, \dots, f_K) \text{ are } \nu\text{-full.}^8$$

Indeed, let $\mathcal{A} := \{\ell \text{ prime} : \prod_{1 \leq j < \nu} \alpha_j(\ell) = 0\}$. Since $\alpha_j(\ell) \geq 1 - (\ell - 1)^{-1} \sum_{i=1}^K \deg F_{i,j}$, we have $\max_{\ell \in \mathcal{A}} \ell \ll 1$. Hence (7.10) would follow once we show that if $(f(n), q) = 1$ and $p^j \parallel n$ with $j < \nu$, then $p \in \mathcal{A}$. Now suppose $p^j \parallel n$ for some $p \notin \mathcal{A}$, $j < \nu$. Then $\prod_{i=1}^K F_{i,j}(p) = f(p^j)$ divides $f(n)$, and $\alpha_j(q) = 0$ so that $\alpha_j(\ell_0) = 0$ for some prime $\ell_0 \mid q$ (Chinese Remainder Theorem). Then $\ell_0 \in \mathcal{A}$, whence $p \in U_{\ell_0}$ (as $p \notin \mathcal{A}$), forcing $\ell_0 \mid \prod_{i=1}^K F_{i,j}(p)$ (as $\alpha_j(\ell_0) = 0$). Hence $\ell_0 \mid f(n)$, contradicting $(f(n), q) = 1$.

The following generalization of the first estimate in Lemma 4.5 will suffice to estimate our input sets.

⁸where we can take the “C” in the definition of ν -full to be some constant depending only on $\{F_{i,j}\}_{\substack{1 \leq i \leq K \\ 1 \leq j \leq \nu}}$.

Proposition 7.7. *As $x \rightarrow \infty$, we have uniformly in ν -admissible $q \leq (\log x)^{K_0}$,*

$$(7.11) \quad \#\{n \leq x : (f(n), q) = 1\} = \frac{x^{1/\nu}}{(\log x)^{1-\alpha_\nu(q)}} \exp(O((\log_2(3q))^{O(1)})).$$

Proof. The lower bound follows by restricting to $n = m^\nu$ and applying Lemma 4.5 to the function $m \mapsto f(m^\nu)$. For the upper bound, let $Y := \exp(\sqrt{\log x})$. By the arguments in the proof of Proposition 7.3, it suffices to bound n of the form $n_1 B A^\nu$ with $n_1 \leq C$, $P(B) \leq Y < P^-(A)$, A squarefree, $B \leq x^{1/2}$ being ν -full, and $(f(B)f(A^\nu), q) = 1$. By [12, Theorem 01, p. 2]⁹, given n_1, B , the number of possible A is $\sum_{A \leq (x/n_1 B)^{1/\nu}} \mu(A)^2 \mathbb{1}_{P^-(A) > Y} \mathbb{1}_{(f(A^\nu), q) = 1} \ll (x/n_1 B)^{1/\nu} (\log x)^{-1} \exp(\sum_{Y < p \leq x} \mathbb{1}_{(f(p^\nu), q) = 1} / p)$. Bounding $\sum B^{-1/\nu}$ as in (7.5), we deduce that the total number of such n is $\ll (x^{1/\nu} / \log x) \exp(\sum_{p \leq x} \mathbb{1}_{(f(p^\nu), q) = 1} / p)$. This is absorbed in the right of (7.11), by the second estimate in Lemma 4.5 applied to $W = \prod_{i=1}^K F_{i,\nu}$. $\square$

Recall $\xi_r(q; z)$ defined in (7.1) (with $h_i := f_i$). Any reference to results or arguments in sections 4-6 are to be understood with $F_{i,\nu}$ playing the role of F_i ; all quantities and parameters are redefined accordingly.

Lemma 7.8. *Fix $\varepsilon > 0$ and assume $K > 1$. If $(F_{i,\nu})_{i=1}^K$ are multiplicatively independent and at least one of them is not squarefull (in $\mathbb{C}$), then $\xi_3(q; z) \ll q^{-2+\varepsilon}$, uniformly in squarefree q and in $z \in (q, x]$.*

Proof. By (4.15) and $\#\mathcal{V}_{3,K}(q; \hat{a}) = \prod_{\ell|q} \#\mathcal{V}_{3,K}(\ell; \hat{a})$, we only need $\#\mathcal{V}_{3,K}(\ell; \hat{a}) \ll \ell$, uniformly in primes $\ell \gg 1$ and $\hat{a} = (a_i)_{i=1}^K \in U_\ell^K$. Now assume without loss of generality that $F_{1,\nu}$ is not squarefull (in $\mathbb{C}$). For all $\ell \gg 1$, we are guaranteed that: **(i)** ℓ doesn't divide the leading coefficients of $F_{1,\nu}$ or $F_{2,\nu}$; **(ii)** $F_{1,\nu}$ is not squarefull over $\overline{\mathbb{F}}_\ell$; and **(iii)** If $F_{1,\nu}^{c_1} \cdot F_{2,\nu}^{c_2}$ is constant in $\overline{\mathbb{F}}_\ell(T)$ for some $c_1, c_2 \in \mathbb{Z}$, then $\ell \mid (c_1, c_2)$.

To ensure (ii), we argue as in the first paragraph of the proof of Lemma 4.3. To ensure (iii), we adapt the proof of Lemma 4.3(2, 3) to see that $(F'_{1,\nu} F_{2,\nu}, F_{1,\nu} F'_{2,\nu})$ must be $\mathbb{F}_\ell$ -linearly independent for all ℓ not dividing the invariant factors of the 2-column matrix listing their coefficients. Now if $F_{1,\nu}^{c_1} F_{2,\nu}^{c_2}$ is constant in $\overline{\mathbb{F}}_\ell(T)$ for any such ℓ , then taking its derivative, we get $c_1 F'_{1,\nu} F_{2,\nu} + c_2 F_{1,\nu} F'_{2,\nu} = 0$, forcing $\ell \mid (c_1, c_2)$.

With $F_i^* := \prod_{j=1}^3 F_{i,\nu}(T_j) - a_i$, we claim that F_1^* is irreducible and $F_1^* \nmid F_2^*$ in $\overline{\mathbb{F}}_\ell[T_1, T_2, T_3]$. Irreducibility is proven as after (4.19). If $F_2^* = F_1^* H^*$ with $H^* \in \overline{\mathbb{F}}_\ell[T_1, T_2, T_3]$, then comparing the terms $T_2^{r_2} T_3^{r_3}$ with maximum $r_2 + r_3$ gives $F_{1,\nu} \mid F_{2,\nu}$, so $F_{2,\nu} = F_{1,\nu} H$ with $m \in \mathbb{Z}^+$, $H \in \overline{\mathbb{F}}_\ell[T]$, $F_{1,\nu} \nmid H$. Thus $F_1^* H^* = F_2^* = (F_1^* + a_1)^m \prod_{j=1}^3 H(T_j) - a_2$, yielding $F_1^* \mid (\prod_{j=1}^3 H(T_j) - a_1^{-m} a_2)$. Write $\prod_{j=1}^3 H(T_j) - a_1^{-m} a_2 = F_1^* Q$; if H were not constant, then comparing the terms $T_2^{r_2} T_3^{r_3}$ with maximal $r_2 + r_3$ gives $F_{1,\nu} \mid H$, a contradiction. Hence $F_{1,\nu}^{-m} F_{2,\nu} = H$ is constant in $\overline{\mathbb{F}}_\ell(T)$, violating (iii) above. This proves $F_1^* \nmid F_2^*$.

In the ring $\mathcal{R} = \overline{\mathbb{F}}_\ell[T_1, T_2, T_3]$, any prime chain $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ containing the ideal $\mathcal{I} = (F_1^*, F_2^*)$ extends to a prime chain $(0) \subsetneq (F_1^*) \subsetneq \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_r$. (This uses irreducibility of F_1^* , that $(F_1^*) \subsetneq (F_1^*, F_2^*)$, and that $\mathcal{R}$ is a UFD.) As $\overline{\mathbb{F}}_\ell[T_1, T_2, T_3]$ has (Krull) dimension 3, we get $r+2 \leq 3$, i.e. $r \leq 1$. By prime correspondence, $\mathcal{R}/\mathcal{I}$ has dimension ≤ 1 , and so does the variety $V_\ell := \{(t_1, t_2, t_3) \in \mathbb{A}_{\overline{\mathbb{F}}_\ell}^3 : (\forall i \in [2]) \prod_{j=1}^3 F_{i,\nu}(t_j) - a_i = 0\}$. Hence, [8, Claim 7.2] yields $\#\mathcal{V}_{3,K}(\ell; \hat{a}) \leq \#V_\ell(\mathbb{F}_\ell) \leq \deg(V_\ell) \ell \ll \ell$ for all $\ell \gg 1$, as desired. $\square$

The entire discussion in § 4.1 now adapts easily, by using Theorem 7.1, Lemma 7.2, and Propositions 7.3 and 7.7 in place of their counterparts in sections 1-4; to estimate $\sum_{p \leq X} \mathbb{1}_{(f(p^\nu), q) = 1} / p$, we use the last estimate in Lemma 4.5 with $W := \prod_{i=1}^K F_{i,\nu}$. Proposition 4.4 holds (for $F_{i,\nu}$), and we use Lemma 7.8 wherever applicable. The analogues of the two inequalities in footnote 7 are also easy to check: For instance, to show that the values of R in (7.8) work, we just need $\max\{s/\nu + r/D, R/\nu - r + r/D - s\} > K$:

⁹This states that for any multiplicative $w : \mathbb{Z}^+ \rightarrow \mathbb{R}_{\geq 0}$ with $\kappa_1 := \sup_{y > 0} y^{-1} \sum_{p \leq y} w(p) \log p$ and $\kappa_2 := \sum_{p, j \geq 2} w(p^j) p^{-j} \log(p^j)$ both finite, we have $\sum_{n \leq x} w(n) \leq (\kappa_1 + \kappa_2 + 1) x (\log x)^{-1} \sum_{n \leq x} w(n)/n$ for all $x > 1$.

If $s/\nu + r/D \leq K$, then $R/\nu - r + r/D - s \geq R/\nu - K\nu + ((\nu + 1)/D - 1)r$, and this last quantity attains its minimum over all $r \in [1, R/\nu]$, either at $r = 1$ or $r = R/\nu$, depending on whether $\nu \geq D$ or $\nu < D$.

Hence, proving Theorems 7.5 and 7.6 comes down to showing that there exists a constant $\delta_0 := \delta_0(C_0, \kappa) < 1$ such that (4.23) holds uniformly in $q \leq (\log x)^{K_0}$ lying in $\mathcal{Q}(\nu; \hat{f})$ having $Q_0 > 1$ and $q^* := \prod_{\ell|q} \ell \gg 1$, and for all nontrivial tuples of characters $\hat{\psi} = (\psi_1, \dots, \psi_K) \bmod Q_0$ with $\prod_{i=1}^K \psi_i(F_{i,\nu}(u))$ being a constant $c_{\hat{\psi}}$ on $\{u \in U_{Q_0} : \prod_{i=1}^K F_{i,\nu}(u) \in U_{Q_0}\}$. We adapt the arguments in section 5 as follows.

7.4. Modification of section 5. In what follows, consider any $x \geq 2$.

Lemma 7.9. *There exists $X \in \left[x, x + \frac{x}{\log^2 x}\right) \cap \mathbb{Z}$ with $\sum_{\substack{3X/4 < n < 5X/4 \\ n \neq X}} \frac{\mathbb{1}_{(f(n), q)=1}}{|\log(X/n)|} \ll X^{1/\nu} \log X$.*

Proof. It suffices to show that $\sum_{X \in \mathbb{Z} \cap [x, x+h)} \sum_{\substack{3X/4 < n < 5X/4 \\ n \neq X}} \frac{\mathbb{1}_{(f(n), q)=1}}{|\log(X/n)|} \ll x^{1/\nu} h \log x$, where $h := \frac{x}{\log^2 x}$.

Write the double sum as $S_1 + S_2$, with S_1 being the contribution of $n \in (3X/4, X - 1]$. Any such $n = X - r$ for some $r \in [1, X/4) \subset [1, (x+h)/4)$. By (7.10), $n = n_1 n_2$ where $n_1 \ll 1$ and n_2 is ν -full. Now $n_2 = (X - r)/n_1 \in [(x - r)/n_1, (x + h - r)/n_1]$, and $|\log(X/n)| = -\log(1 - r/X) \gg r/x$. Hence,

$$S_1 \leq x \sum_{1 \leq r \leq \frac{x+h}{4}} \frac{1}{r} \sum_{\substack{n_1 \ll 1 \\ \frac{x-r}{n_1} \leq n_2 < \frac{x+h-r}{n_1} \\ n_2 \text{ is } \nu\text{-full}}} 1 \ll x \log x \left(x^{1/\nu} \cdot \frac{h}{x} + x^{1/(\nu+1)} \right) \ll x^{1/\nu} h \log x,$$

where we used Erdős-Szekeres [9] to estimate the sum on n_2 . Likewise, $S_2 \ll x^{1/\nu} h \log x$, as desired. $\square$

As $\sum_{x < n \leq x+x/\log^2 x} \mathbb{1}_{(f(n), q)=1} \leq \sum_{n_1 \leq C} \sum_{n_1^{-1}x < n_2 \leq n_1^{-1}(x+x/\log^2 x)} \mathbb{1}_{n_2 \text{ is } \nu\text{-full}} \ll x^{1/\nu} / \log^2 x$, it suffices to show (4.23) with x replaced by the X in Lemma 7.9. The series $\mathcal{F}(s) = \sum_{n \geq 1} \mathbb{1}_{(f(n), q)=1} \hat{\psi}(\hat{f}(n))/n^s$ converges absolutely on $\text{Re}(s) > 1/\nu$, as all but $O(1)$ -many Euler factors are $1 + \sum_{r \geq \nu} \mathbb{1}_{(f(p^r), q)=1}/p^{rs}$ by ν -admissibility. Hence we may apply Perron on the vertical line $\text{Re}(s) = \nu^{-1}(1 + 1/\log X)$ to get the analogue of (5.1); Lemma 7.9 bounds the contribution of $n \in (3X/4, 5X/4)$ in the corresponding error term, while that of the other n is bounded by writing $n = n_1 n_2$ and treating the ν -full n_2 as in (5.2).

The rest of the argument in section 5 goes through by scaling things by ν appropriately, and with $F_{i,\nu}$ replacing F_i . Branch cuts on $(-\infty, 1]$, $(-\infty, \rho_e]$ are replaced by $(-\infty, 1/\nu]$, $(-\infty, \rho_e/\nu]$. Redefining $\mathcal{L}_{q^*}(t) = \log(q^*(|t\nu| + 1))$, Lemma 5.1 holds with $F_{\hat{\psi}}(s\nu) := \left(\prod_{\eta \bmod Q} L(s\nu, \eta)^{\gamma(\eta)}\right)^{\alpha_{\nu}(Q)c_{\hat{\psi}}}$ replacing $F_{\hat{\psi}}(s)$, and with $\gamma(\eta)$, $G_{\hat{\psi}}$ redefined accordingly; the new $G_{\hat{\psi}}$ is analytic on $\sigma > \nu^{-1}(1 - 1/\log q^*)$ and satisfies $G_{\hat{\psi}}(s) \ll (\log_2 q^*)^3$ uniformly therein. Contour Γ_1 is redefined by scaling $\vartheta(t)$, $1 + 1/\log x$, 1 , $\tilde{\rho}_e, \rho_e$ down by ν (i.e. now $\vartheta(t) := \nu^{-1}(1 - c_1/4\mathcal{L}_{q^*}(t))$ etc.), but we can still take $T = \exp(\sqrt{\log x})$.

We redefine $H_{\hat{\psi}}(s)$ to $H_{\hat{\psi}}(s) := F_{\hat{\psi}}(s\nu) (s - 1/\nu)^{\alpha_{\nu}(q)c_{\hat{\psi}}} (s - \rho_e/\nu)^{-\alpha_{\nu}(q)\gamma(\eta_e)c_{\hat{\psi}}}$ which analytically continues into $\sigma > \nu^{-1}(1 - c_1/\mathcal{L}_{q^*}(t))$. The analogue of (5.7) holds with “ $1 - c_1/2\mathcal{L}_{q^*}(t) \leq \text{Re}(s) \leq \dots$ ” replaced by “ $\nu^{-1}(1 - c_1/2\mathcal{L}_{q^*}(t)) \leq \text{Re}(s) \leq \dots$ ”. Proposition 5.3(2) holds for $s \in [\nu^{-1}(1 - c_1/4\log q^*), \nu^{-1}]$; in the proof of subpart (1), we redefine $\mu(t) := \nu^{-1}(1 + c_1/7\mathcal{L}_{q^*}(t))$, and replace $\zeta(\mu(t))$ by $\zeta(\nu\mu(t))$.

Towards the end (in the case $c_{\hat{\psi}} = 1$), we take $G_{\hat{\psi},2}(s) := \prod_{p \leq 2\nu} (1 + \sum_{r \geq 1} \mathbb{1}_{(f(p^r), q)=1}/p^{rs})$, and write $G_{\hat{\psi}}(s) = G_{\hat{\psi},1}(s)G_{\hat{\psi},2}(s)$, where $G_{\hat{\psi},1}(s)$ is analytic on $\sigma > \nu^{-1}(1 - 1/\log q^*)$ and satisfies $G_{\hat{\psi},1}(s) \ll (\log_2 q^*)^3$ uniformly therein. By (7.7) and its ensuing observation, $G_{\hat{\psi},2}(1/\nu) = 0$. This proves (4.23). $\square$

7.5. Optimality in Theorems 7.5 and 7.6. Fix any $\nu \in \mathbb{Z}^+$. We will construct $\widehat{F} := (F_{1,\nu}, \dots, F_{K,\nu})$ with $\prod_{i=1}^K F_{i,\nu}$ squarefree. The arguments for Lemma 6.1 show that there exists a constant $C_{\widehat{F}} > 0$ such that any q with $P^-(q) > C_{\widehat{F}}$ has $\alpha_\nu(q) \neq 0$ and satisfies (7.7) vacuously (there is *no* tuple of characters $\widehat{\chi} \neq (\chi_0, \dots, \chi_0) \pmod q$ with $\prod_{i=1}^K \chi_i(F_{i,\nu}(u)) = 1$ on $\{u \in U_q : \prod_{i=1}^K F_{i,\nu}(u) \in U_q\}$). We fix a prime $\ell_0 > C_{\widehat{F}}$, and fix $\{F_{i,j}\}_{\substack{1 \leq i \leq K \\ 1 \leq j \leq \nu-1}} \subset \mathbb{Z}[T]$ with **all** coefficients divisible by ℓ_0 . Our moduli $q \leq (\log x)^{K_0}$ will have $P^-(q) = \ell_0$, hence we may take $\ell_0 \gg_{F_{1,\nu}, \dots, F_{K,\nu}} 1$ so that the respective lower bound conditions on $P^-(q)$ in section 6 are satisfied. The discussion in this paragraph then guarantees that $q \in \mathcal{Q}(\nu; \widehat{f})$.

Now, all constructions before “*Optimality in Theorem 1.3*” in section 6 can be adapted by replacing F_i by $F_{i,\nu}$; the lower bounds come from inputs $n = P^\nu$ rather than $n = P$. For instance, to prove optimality in Theorem 7.5(i), take $F_{i,\nu}(T) := (T-1)^r + i$, and q a perfect r -th power with $P^-(q) = \ell_0 > \max\{C_{\widehat{F}}, 2K\}$, so that $\#\{n \leq x : (\forall i) f_i(n) \equiv i \pmod q\} \gg \#\{P : P^\nu \leq x, P \equiv 1 \pmod{q^{1/r}}\} \gg x^{1/\nu} / (q^{1/r} \log x)$.

To show the optimality of $R = \nu(K\nu + K - \nu) + 1$ in (7.9) (for q squarefree), we adapt the last construction in section 6: Take $F_{i,\nu}(T) := \prod_{j=1}^r (T-2j) + 2(2i-1)$, and continue with the set-up in the first paragraph of § 7.5 (with $P^-(q) = \ell_0 > \max\{C_{\widehat{F}}, 4Kr\}$). Define $f_i(p^j) = F_{i,j}(p)$ and $f_i(p^{\nu+1}) = 1$ for all $i \in [K]$, $j \in [\nu]$. Take $n = (p_1 \dots p_{(K-1)\nu})^{\nu+1} P^\nu \leq x$, with the primes p_i, P satisfying $q < p_{K-1} < \dots < p_1 < x^{1/8K\nu^2} < x^{1/3\nu} < P$ and $\prod_{j=1}^r (P-2j) \equiv 0 \pmod q$. Proceeding as in the rest of section 6 thus yields

$$\sum_{\substack{n \leq x: P_{\nu(K\nu+K-\nu)}(n) > q \\ (\forall i) f_i(n) \equiv 2(2i-1) \pmod q}} 1 \gg \frac{r^{\omega(q)}}{\varphi(q)^K} \cdot \frac{x^{1/\nu}}{(\log_2 x)^{\nu(K-1)+1} \log x}$$

which, just like the right of (6.3), can be made to grow much faster than the expected main term.

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